Branches and roots: bank presence and forest preservation in Amazonian Indigenous lands*

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Abstract

Indigenous lands shelter a large portion of the Brazilian Amazon's intact forest, yet illegal deforestation inside their boundaries has been climbing despite legal protection. We exploit the expansion and subsequent contraction of the bank-branch network between 2008 and 2024 in the traditionally underbanked Amazon to analyze whether local banking presence affects deforestation inside nearby Indigenous lands. Using branch dynamics in surrounding municipalities as instrumental variables, we find that each additional branch near an Indigenous land lowers cumulative forest loss inside that area by roughly 2.7 km² within three years. We then show that this effect runs through local social capital. A single-instrument mediation design separates the mediation that occurs through social capital from the mediation that does not run through it. Branches raise social capital, social capital lowers deforestation with branch presence held fixed, while the effect running through the remaining channels is negligible. New branches also bring more restaurants, bars and supermarkets, the everyday venues where social ties form. The results link the evidence that trust underpins finance to the evidence that social capital sustains public goods and deters crime.

Keywords: Deforestation, banks, Indigenous lands, instrumental variables, mediation analysis.

JEL codes: Q56, G21, Q23, O13, R11.

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1 Introduction

Indigenous lands serve as effective barriers to deforestation in the Brazilian Legal Amazon (BLA), where they cover roughly one quarter of the total area.[1] Their protective role is supported by Indigenous territorial governance and forest stewardship, alongside legal restrictions on resource exploitation by outsiders (Fetzer and Marden, 2017; Amin et al., 2019; Walker et al., 2020; Sze et al., 2022; Qin et al., 2023; Virtanen et al., 2025). However, these forest sanctuaries suffer growing deforestation, primarily associated with criminal activities such as illegal mining (Silva-Junior et al., 2023). Since deforestation requires the presence of perpetrators and heavy machinery on site, local banks can plausibly shape incentives and capacity for this activity; entry and exit of bank branches can be particularly consequential in those remote, historically underbanked communities (Beck et al., 2010; Fonseca and Matray, 2024). The pronounced changes in banking presence in the BLA in the last two decades therefore raise the question of whether and how banks affect Indigenous land deforestation.

We estimate this effect causally across 381 Indigenous lands, combining instrumental variables with granular administrative data. We instrument the endogenous change in branch presence in the surrounding municipalities with the corresponding change in an *outer ring* that does not border the land. Relevance comes from both rings responding to the same regional banking dynamics. Identification assumes that outer-ring changes affect deforestation inside the Indigenous land only through their effect on branches in those municipalities. We then use a single-instrument mediation analysis (Dippel et al., 2021; Heckman and Pinto, 2024) to ask whether the effect runs through local social capital. Our sample period, 2008 to 2024, spans federal administrations with contrasting environmental policies and three full business cycles in Brazil.

Our first main finding is that each additional branch neighboring an Indigenous land reduces its cumulative deforestation by about 2.7 km^2 within three years, an effect that persists through at least five years. This protective effect is equivalent to about 4 years of the average land's annual forest loss. Since bank branches cannot physically prevent forest clearing, this effect must result from indirect channels in the surrounding communities brought about by these new branches. Building on evidence that social capital sustains common resources and deters crime (Ostrom, 1990; Sampson et al., 1997; Rustagi et al., 2010), and that financial institutions reshape the ties among those they serve (Feigenberg et al., 2013), we aim the mediation analysis not at credit but at the social fabric (Bourdieu, 1983) of these small communities, sensitive to the arrival of a single branch. Specifically, we estimate how much of the total effect runs through social capital and how much does not.

This mediation analysis yields our second main finding: the protective effect of bank presence on forests within Indigenous lands runs through the increase in local social capital that branch entry produces. We

[1] Collectively, these Indigenous lands correspond to an area of 1,158,701 km^2, or approximately three times the area of Germany.

proxy this mediator with the first principal component of the valid vote share and non-profit association employment, following Rupasingha et al. (2006) and adapting the proxies of Guiso et al. (2004) to the Brazilian context. Identification rests on mediated confounding (Dippel et al., 2021; Heckman and Pinto, 2024): the unobserved factors that drive branch placement may move social capital, but they do not drive deforestation directly, while unobserved confounding may exist between social capital and deforestation. Under this assumption, bank presence raises our measure of social capital, and social capital in turn lowers deforestation. The mediation that does not run through social capital is small and statistically indistinguishable from zero at every horizon. The effect is concentrated where FUNAI, the federal agency for Indigenous peoples, has an office nearby, which suggests that it works alongside formal enforcement rather than replacing it.

The paper sits where finance, environmental and resource economics, development, and the study of social capital meet. Each in turn has measured how credit reaches the forest frontier, how protected areas and enforcement slow clearing, how bank branches change local economies, and how civic norms promote finance and sustain public goods. The link we document runs across all four: a financial institution changes the civic life of a town, and the town's civic life protects a forest it does not own.

First, we contribute to the literature on finance and deforestation. The economics of tropical deforestation has centred on the returns to clearing and the enforcement that offsets them (Balboni et al., 2023), and within it finance has mostly been studied as a force that clears forest. Rural credit funds the expansion of pasture (Assunção et al., 2020), and banks shift agribusiness lending towards deforestable regions when enforcement weakens (Berger et al., 2026). Branches also change the places they serve, raising local economic activity (Burgess and Pande, 2005; Beck et al., 2010; Bruhn and Love, 2014; Fonseca and Matray, 2024); in the Amazon much of that presence is state-owned, with state banks reaching the deforesting frontier (Ortega and Ossandon Busch, 2025). We study a setting where clearing is illegal and the forest is held in common, and find the opposite sign: banks protect the forest, and they do so through presence rather than credit.

Second, we contribute to the study of Indigenous lands as instruments of conservation. Their protective effect is well documented (Blackman and Veit, 2018; Walker et al., 2020; Qin et al., 2023), and recent work identifies roads, fires and illegal mining as the proximate drivers of clearing inside them (Da Silva et al., 2023a,b), while the political economy of enforcement shapes how much clearing those drivers produce (Burgess et al., 2012). We show that formal banking access in the towns around an Indigenous land shapes how much of it is cleared, making the communities around the territory part of how it is protected.

Third, we contribute to the literature on social capital. Its consequences for financial development, crime and the management of common resources are well established (Ostrom, 1990; Sampson et al., 1997; Guiso et al., 2004; Rustagi et al., 2010). Its determinants, especially those open to policy, are less understood (Glaeser et al., 2002; Durlauf and Fafchamps, 2005). One exception is Feigenberg et al. (2013), who show

experimentally that repeated meetings of microfinance groups build lasting social ties. We show that a bank branch, a far more common institution, does the same in small and dispersed Amazonian towns, and that the social capital it builds has an environmental return.

Finally, we contribute to causal mediation with a single instrument. Dippel et al. (2021) show that one instrument identifies the effect of a mediator when the unobservables behind the treatment are independent of those behind the outcome. We show what the coefficient on branch presence captures when the effect runs through any number of unmodeled mediators. If branches have no direct effect, that coefficient measures exactly the mediation that does not run through social capital. Our approach complements tests of full mediation that do not require mediated confounding but need a discrete mediator and a binary, as-good-as-random treatment (Kwon and Roth, 2026). Our instrument is spatial, and we assess it following recent work on spatial treatments (Pollmann, 2026), with placebo instruments built from farther rings of municipalities and with design-based randomization inference.

These findings matter for policy. Bank branches in remote areas are closing as banking moves online, and the policy debate weighs these closures against the cost of serving sparse populations or those with limited internet access. Our results suggest that the closures also remove a community institution, with measurable consequences for forest protection that the current debate overlooks.

The next section offers background context about Indigenous lands in the BLA and developments in banking in the region. Section 3 sets out a model of the incentives behind illegal deforestation and the place of social capital within it. We then describe in Section 4 the granular administrative datasets that we use in this paper, including the unit of analysis on which our empirical strategy rests.[2] Section 5 presents our results connecting bank presence to deforestation in Indigenous lands, and then builds on them with mediation analyses that identify the causal channels, including how bank presence raises social capital. Section 6 probes the exclusion restriction and reports further checks on the sample, the inference, and heterogeneity. Section 7 concludes.

2 Institutional context

2.1 Indigenous lands and deforestation

According to the United Nations, nearly half a billion Indigenous people live in 90 countries worldwide. These groups are often socially marginalized and face threats to their land rights. The main legally binding international instrument addressing these challenges is the Indigenous and Tribal Peoples Convention of 1989 (ILO Convention No. 169), which obliges ratifying states to recognize and protect Indigenous peoples'

[2]All data used in this paper is publicly available, and all analysis was conducted with open source software. We provide the code for the full replication of our analysis upon request.

ownership of the lands they traditionally occupy (International Labour Organization, 1989). The Constitution of Brazil, one of the 24 countries that have ratified ILO 169, explicitly acknowledges Indigenous peoples' rights over their lands and charges the federal government with demarcating and protecting those territories and their resources.

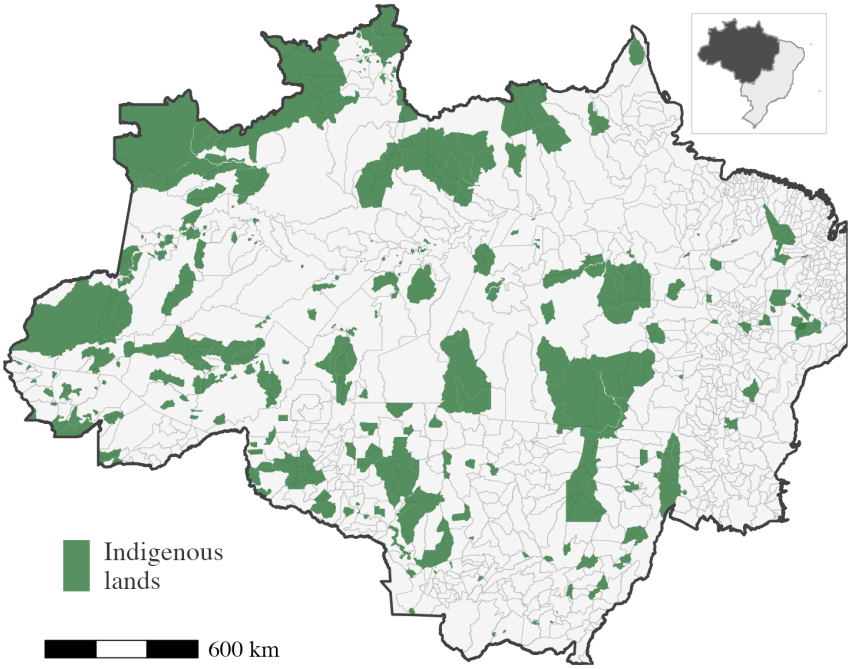


Figure 1: Spatial distribution of Indigenous lands across the Brazilian Legal Amazon. Thin lines represent municipality borders. The inset shows the region covered by the main map (dark grey) within Brazil (light grey). Source: FUNAI (Indigenous lands) and IBGE (municipal boundaries).

As of 2024, the Brazilian government had officially delimited 626 Indigenous lands nationwide (World Bank, 2025); 381 of these are located in the BLA, covering approximately 25% of that region's total area. Figure 1 illustrates the geographic distribution of these Indigenous lands. Together, these territories are home to at least 421,655 people (Instituto Brasileiro de Geografia e Estatística, 2022) of at least 319 distinct ethnic groups and speaking 157 different languages (ISA, 2026).

Although demarcation exists to secure these peoples' permanent possession of the lands they traditionally occupy, these territories have also functioned as effective barriers to the expansion of deforestation in the Amazon rainforest (Blackman and Veit, 2018). Walker et al. (2020) show that Indigenous lands suffered 63% lower carbon stock losses compared to the rest of the Amazon between 2003 and 2016. Related evidence indicates that demarcation is also linked to reduced deforestation in areas surrounding Indigenous lands (De Assis Barros et al., 2022). These outcomes reflect a combination of governance and institutional

mechanisms. Indigenous governance and forest stewardship structures tend to align long-run incentives with forest conservation (Virtanen et al., 2025). In parallel, non-Indigenous economic activities within Indigenous lands are strictly regulated and must be explicitly and individually allowed. Enforcement is formally assigned to the National Foundation of Indigenous Peoples (FUNAI) and other environmental and law enforcement agencies, including at the subnational level.[3]

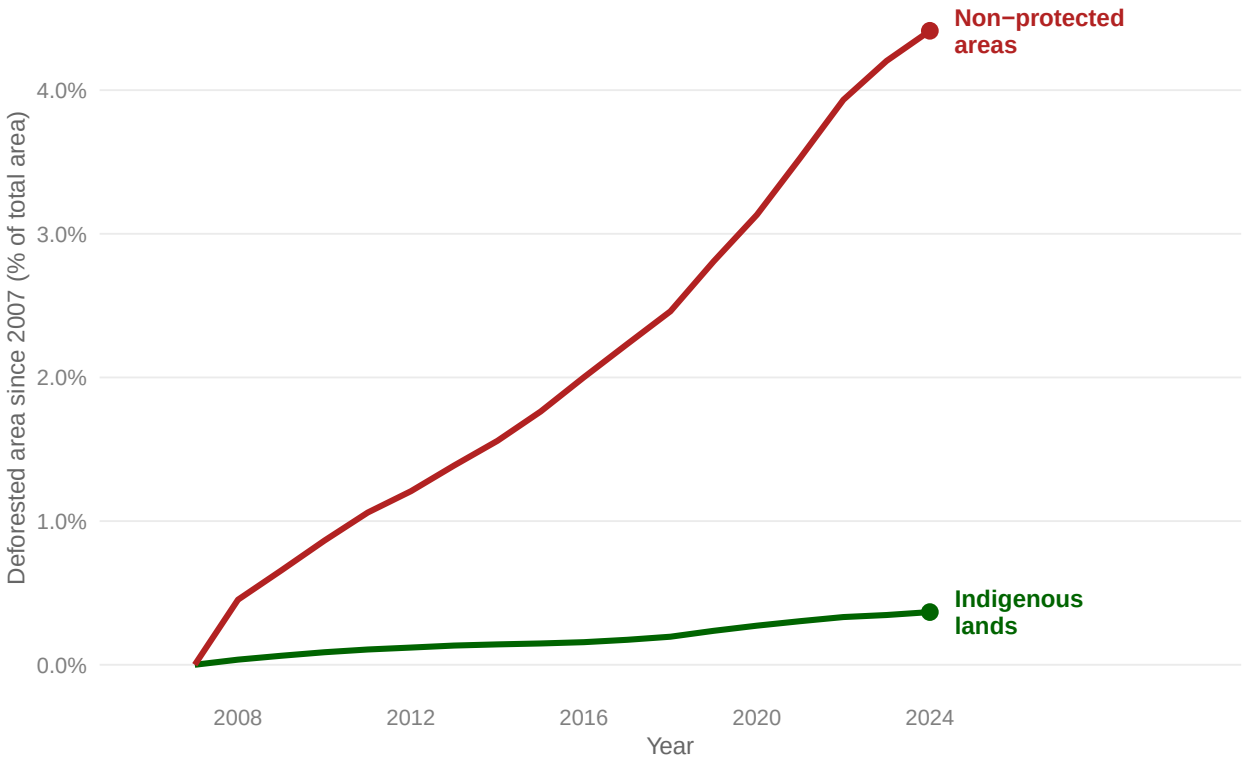


Figure 2: Cumulative deforestation: Indigenous lands vs non-protected areas.

However, despite their protective role, deforestation inside Indigenous lands does occur and is on the rise. Figure 2 compares deforestation trends inside Indigenous lands with those in non-protected areas of the Legal Amazon, showing that forest loss in Indigenous lands is lower than that in non-protected lands but steadily increasing.[4] The main drivers of deforestation in these areas include illegal mining, clandestine airstrips, and the expansion of unofficial roads that facilitate access for extractive activities (Da Silva et al., 2023b,a).

[3]For example, the vast majority of public works/construction inside Indigenous lands approved by FUNAI are new housing, building or updating energy or water treatment infrastructure, and the build-up of health units.

[4]Small as that share is, it understates the harm: the underlying illegal activity brings armed intrusion, exposes Indigenous populations to imported disease, and carries environmental externalities such as mercury contamination from artisanal gold mining (Parfitt, 2025).

2.2 Trends in banking in the Brazilian Amazon

Branch presence in the BLA over our sample period of 2008 to 2024 exhibited strong variation over time. The regional branch stock expanded rapidly during the post-GFC commodity boom and a nationwide policy to expand the banking network, reached a peak around 2015, and then declined steadily thereafter (Figure 3). This inflection aligns with the rapid diffusion of mobile banking, a shift that has been particularly pronounced in Brazil.

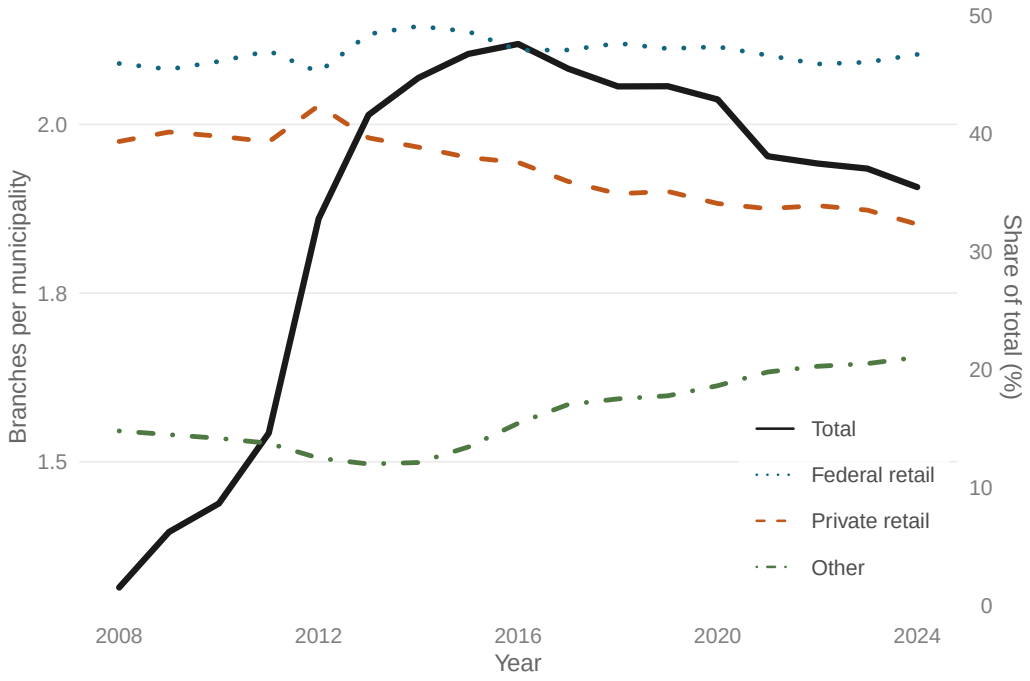


Figure 3: Bank branches per municipality near Indigenous lands, by ownership group. Solid line: average branches per municipality across all banks (left axis). Dashed lines: each group's share of the total (right axis). Federal retail is Banco do Brasil and Caixa Econômica; Other groups federal development banks, the Pará state bank, and non-bank institutions.

The analysis that follows concentrates on the municipalities that contain or directly border an Indigenous land, since these are the places whose bank branches are physically within reach of the territory; Section 4 defines this set precisely. Banking presence there is predominantly government-owned: these banks account for 63% of branch presence in our sample period, measured in branch-years, against 37% for private sector banks. Government-owned banks span the federally controlled retail banks Banco do Brasil and Caixa Econômica Federal (or "Caixa" for short), the federal development banks Banco da Amazônia and Banco do Nordeste, and the Pará state bank Banpará. Banco do Brasil is the single largest presence, ahead of second-place Bradesco, with Caixa third. Figure 3 traces the average number of branches per such municipality, together with the share held by each ownership group.

Institution	Share of variation (%)
Federal retail banks	**39.9**
Banco do Brasil	27.4
Caixa Econômica	12.5
Private retail banks	**45.4**
Bradesco	29.7
Itaú	5.1
Santander	4.2
HSBC	5.6
Other private banks	0.7
Development, state, and non-bank institutions	**14.7**
Banpará	7.4
Banco da Amazônia	5.5
Banco do Nordeste	0.4
Non-bank institutions	1.3

Table 1: Share of bank-branch variation by institution, 2008 to 2024. Branch variation is the sum of absolute year-on-year changes in each bank's branch count across Indigenous-land municipalities; the table reports each institution's share of the total.

While total banking presence in the BLA oscillated markedly, the composition remained broadly stable over time, with federal retail banks holding close to half of the branches throughout. The private-retail share receded modestly after the mid-decade peak, while the share of other institutions rose slightly. That stability masks considerable turnover at the branch level, with branches opening and closing in every year even as the ownership shares hold roughly constant. Table 1 decomposes this movement, measured as the sum of each institution's year-on-year branch changes across the sample. Although privately owned banks are a minority presence, they account for close to half of all branch variation, about 45%. Their adjustment is two-sided: the largest expansions came without offsetting closures, as Caixa and Banpará opened branches without closing any, whereas the private banks both opened and closed them, HSBC's 2016 exit from Brazil being the sharpest case.

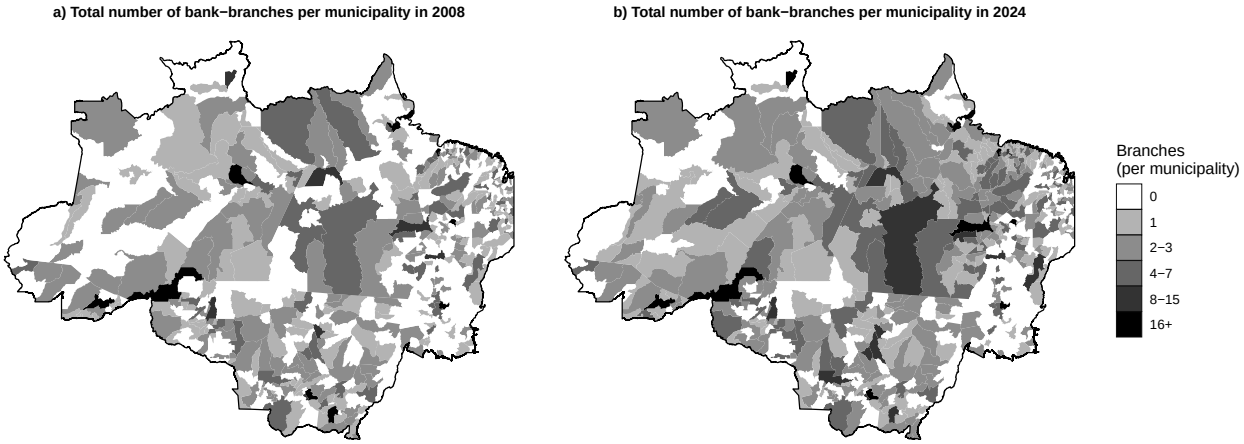


Figure 4: Bank-branch coverage across Legal-Amazon municipalities: a) 2008 vs. b) 2024

The spatial configuration of commercial banking in the BLA also changed considerably in our sample period. Figure 4 contrasts branch density across municipalities at the beginning and end of the sample. In 2008, 420 municipalities, corresponding to 54% of those in the study region, had no physical bank branch. By 2024, this number had fallen to 346 (45%).

3 Conceptual framework

3.1 A model of illegal deforestation incentives

We connect banking presence to illegal deforestation inside Indigenous lands with a simple model, which we then use to inform our empirical strategy. Illegal clearing inside Indigenous lands is the result of an active choice made by outsiders who enter these territories with the necessary heavy machinery to operate. Perpetrators weigh what the cleared land (or another criminal activity that requires forest clearing, such as goldmining) would earn against potential earnings from legitimate work, the chance of being caught and the social cost that comes with the choice of criminal activity (Becker, 1968). We write that choice down in a deliberately simple model, whose purpose is to highlight how banks can reach deforestation through social capital, and to show what the mediation analysis must hold fixed (Heckman and Pinto, 2015, 2024; Huber, 2020).

The model is stated for a single Indigenous land and the municipalities around it, indexed by i, with time subscripts suppressed. Unlike most crimes, clearing forest requires substantial capital. What it earns follows a Cobb-Douglas technology in criminal labor L_i, with share α, and physical capital K_i such as heavy machinery, net of logistical costs ℓ_i. This function covers timber, cattle pasture and mining alike. Against that payoff perpetrators weigh the disutility of being caught, J, with the two outcomes weighted by the

probability of capture p_i. Subtracting the utility of legitimate work, $u(y_i)$, which depends on local economic conditions C_i, gives

$$U_i = (1 - p_i)u(L_i^{\alpha} K_i^{1-\alpha} - \ell_i) + p_i u(J) - u(y_i), \tag{1}$$

The probability of being caught, p_i, lies between zero and one and depends on enforcement capacity E_i, a common unobserved confounder V, and independent shocks. Enforcement capacity summarizes the resources the state can bring to monitoring and protecting the land. Higher enforcement raises the chance of capture and so lowers the expected return to clearing. An active literature exploits variation in enforcement (Assunção et al., 2023; Merkus, 2024; Berger et al., 2026). The level of punishment J is held fixed throughout.

Perpetrators compare that payoff against a reservation utility $\underline{U}_i$, the threshold the payoff must exceed before they engage in clearing. Since the wage from legitimate work is already netted out in U_i, what $\underline{U}_i$ carries is social rather than financial, and it depends in part on local social capital. This is the social cost of offending, or the social capital forfeited when perpetrators choose a criminal activity; Rasmusen (1996) models it as the stigma attached to a criminal reputation. Higher social capital raises this threshold, so clearing must pay more to be worth it. Social sanctions and formal enforcement both deter clearing, and the two can relate in opposite ways. They are substitutes if communities sanction offenders where the state does not (Guiso et al., 2004), and complements if communities protect the forest by supporting the agencies that enforce the law. Section 6.6 tests which of the two holds. Section 3.2 develops the concept further.

Bank presence can influence several of these objects (Fonseca and Matray, 2024; Fang and Vlaicu, 2024), and they can influence each other. For example, social capital can strengthen enforcement if residents report illegal clearing to the authorities. It also co-varies with economic activity, since the restaurants and bars that local credit finances are where residents meet. Branches can therefore foster social capital both directly, by drawing dispersed households into town, and indirectly, by financing the venues in which they meet (Putnam et al., 1993; Mayer et al., 2022). Other routes do not run through social capital at all. For example, bank-financed heavy machinery can change local economic conditions without passing through social capital. Our focus is on the aggregate mediation that runs through social capital. Every route from branches to deforestation either passes through social capital or does not, whatever else it involves (Figure 5). While our focus is on mediation through social capital, we do not claim it is the only route, and the estimates allow for others.

In the framework, social capital raises the threshold and so makes clearing less likely. The mediation that does not run through social capital can push in either direction. For this reason, we estimate its combined contribution rather than predicting its sign. Section 5.3 estimates both in the same step.

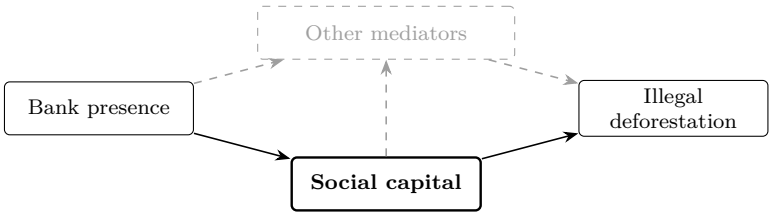


Figure 5: Bank presence reaches deforestation through mediation that occurs through social capital (solid) and mediation that does not run through social capital (dashed). The mediation analysis estimates the first and accounts for the second. Mediators connected to social capital count towards the first along the paths that run through it.

3.2 Social capital: concept and measurement

Social capital refers to the features of social structure, such as social bonds and norms, that facilitate collective action (Coleman, 1988). Communities with a higher stock of social capital find cooperation easier, as trust reduces the frictions inherent in most economic transactions (Putnam et al., 1993). For empirical purposes, the precise definition matters little, as different measures of social capital tend to be highly correlated in practice (Guiso et al., 2004). Guiso et al. (2011) narrow the concept to the values and beliefs that help a group cooperate, which they call civic capital.

Social capital sits between finance and the commons. On the finance side, trust is embedded in every exchange (Arrow, 1972) and civic norms raise economic performance (Knack and Keefer, 1997). Households in high-trust regions use more checks, less cash and more formal credit (Guiso et al., 2004), and hold more stocks (Guiso et al., 2008). Trust predicts loan repayment (Karlan, 2005), social networks serve as collateral (Karlan et al., 2009), and cultural proximity between loan officers and borrowers raises both (Fisman et al., 2017; Hasan et al., 2022). The reverse direction, from financial institutions to social capital, has received much less attention. Feigenberg et al. (2013) show experimentally that microfinance clients randomly assigned to meet weekly rather than monthly form lasting social ties, pool risk more and default less. Social connections fall sharply with physical distance (Glaeser et al., 2002), so in a sparsely populated region a branch that draws dispersed households into town can create interaction that would not otherwise happen. The cafés, bars and shops that local credit sustains are the "third places" where community life takes place (Oldenburg, 1989). This paper studies that reverse direction.

In our setting, Indigenous lands are a common-pool resource in the sense of Ostrom (1990): excluding illegal actors is costly and incomplete. Collective governance and vigilance depend on shared norms of reciprocity and trust. Auer et al. (2020) argue that social capital formation is a valuable conservation tool, particularly for Indigenous communities, where power asymmetries would otherwise lead to their exclusion from resource governance. Communities can sustain such resources through shared norms and mutual monitoring (Ostrom, 2000; Pretty, 2003), and forest user groups with more conditional cooperators monitor

more and keep better forests (Rustagi et al., 2010). Evidence from the Brazilian Amazon points the same way: Mertens et al. (2011) show that social capital among local actors predicts the success of collective environmental governance. Social capital also makes institutions more accountable, since voters punish misbehaving politicians more where it is high (Nannicini et al., 2013).

Higher social capital is associated with lower crime rates (Sampson et al., 1997; Lederman et al., 2002; Buonanno et al., 2009; Akçomak and ter Weel, 2012). This operates through peer sanctioning and community norms. Glaeser et al. (1996) show that criminal behavior responds to local social interactions, and social capital raises the social cost of defection by making deviance visible and punishable within the community (Rasmusen, 1996). Since illegal deforestation is itself a criminal activity, this mechanism applies directly to our setting. Where formal enforcement is weak, social capital has been found to matter more. Guiso et al. (2004) show that the effect of social capital on economic outcomes is strongest in regions with less reliable legal institutions, as may be the case in parts of the BLA. Social capital is not protective by construction, however: dense civic networks can also serve harmful ends (Satyanath et al., 2017). Our estimates therefore recover the sign of its effect on deforestation rather than assume it.

Measuring social capital directly is difficult, so the literature relies on observable civic behaviors as proxies (Glaeser et al., 2000; Durlauf and Fafchamps, 2005).[5] The most common are electoral participation and civil society membership (Putnam et al., 1993; Guiso et al., 2004; Rupasingha et al., 2006; Buonanno et al., 2009; Akçomak and ter Weel, 2012). Since these indicators capture different facets of the same latent construct and tend to be highly correlated, several papers aggregate them into a single index using principal component analysis (Rupasingha et al., 2006; Akçomak and ter Weel, 2012; Guriev and Melnikov, 2016). We follow this approach, constructing a social capital index (SCI) as the first principal component of two inputs: the valid vote share and non-profit association employment per 100,000 inhabitants, both at the ILTF level.[6]

Elections in Brazil are biennial, so in each odd year we use the valid vote share of the preceding election. The principal component is computed separately within each ILTF and signed so that it loads positively on the valid vote share. Results for the individual proxies are reported in Appendix C. Within Indigenous lands the index is essentially uncorrelated with local economic conditions: its correlation with GDP, exports, rural credit, tax revenue, vehicles, population and urban expansion is near zero in every case (Table A3.8). We use the valid vote share instead of turnout because voting is mandatory in Brazil. Turnout then reflects the cost of reaching the polling station as much as civic engagement, while a voter already at the booth is still free to cast a blank or void ballot, or to vote for a non-existent candidate. A blank or void ballot is informative precisely because it is a choice, not an obstacle.

[5] Another strategy to collect indicator data on social capital is to conduct surveys, usually on questions about mutual trust (Knack and Keefer, 1997; Glaeser et al., 2000; Alesina and La Ferrara, 2002). No such survey exists with the granularity and time coverage our setting requires.

[6] Variables scaled by population (e.g., association employment) use the 2022 IBGE census population for 2022, 2023, and 2024, since municipal-level population data for 2023 and 2024 were not available at the time of writing.

4 Data

4.1 Unit of analysis: the Indigenous land territorial footprint (ILTF)

While some Indigenous lands are entirely contained within a single municipality, others extend across several. Their legal status and governance, however, are independent of municipal boundaries. Local bank branches, by contrast, are located in the urban centers of the surrounding municipalities, most of which are small towns. To capture the effect of banking presence in the municipalities with direct access to a given Indigenous land, we define the Indigenous Land Territorial Footprint (ILTF) as our unit of analysis. The ILTF of each Indigenous land is the set of municipalities that intersect the land or lie within 1 km of its boundary.

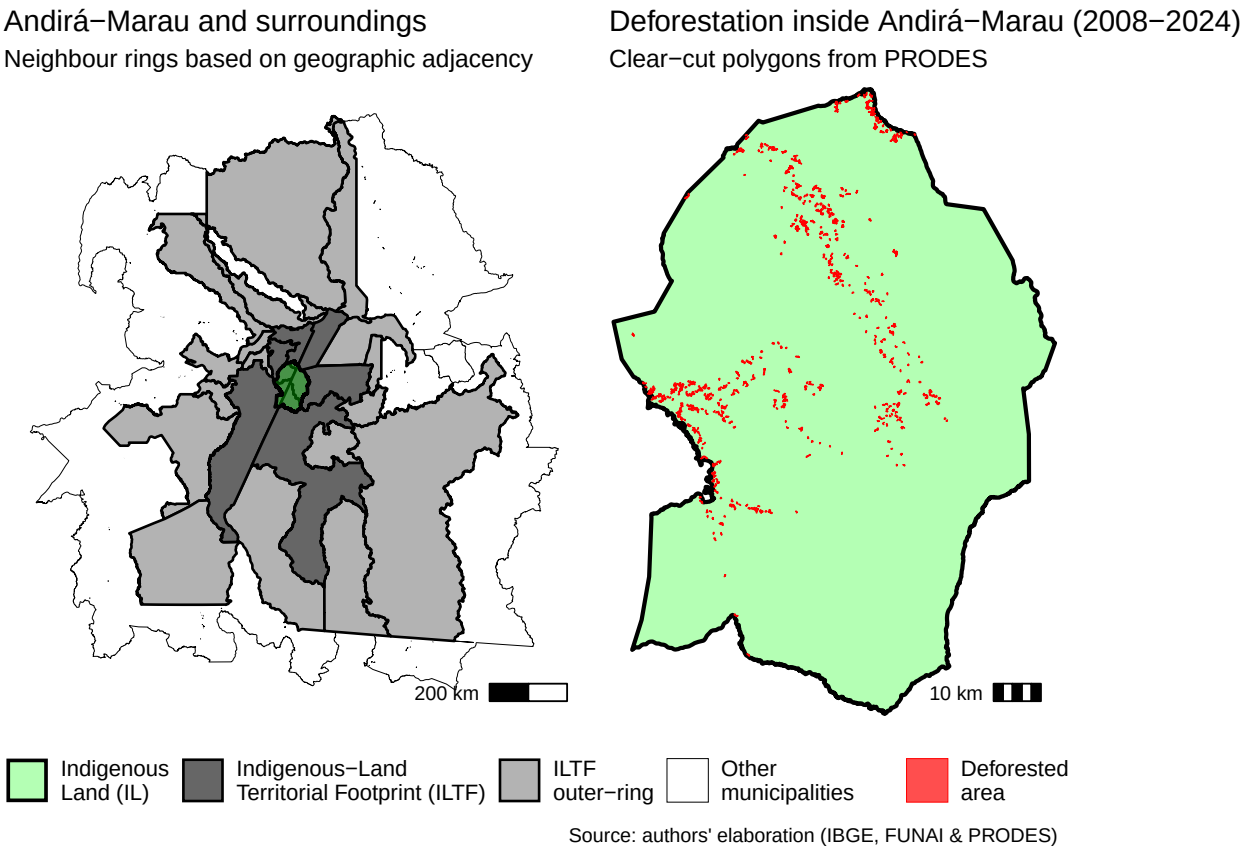


Figure 6: Left: Andirá-Marau Indigenous Land and neighboring municipality rings. Right: Clear-cut deforestation polygons inside the IL (2008–2024, PRODES imagery).

Figure 6 illustrates the concept using the *Andirá-Marau* Indigenous land as an example. The left panel maps the municipality rings: the central green area is the Indigenous land, the surrounding dark-gray area is its ILTF, and the light-gray municipalities form the outer ring, defined as the first-order neighbors of the ILTF that border it but do not touch the Indigenous land itself. Branch dynamics in this outer ring

serve as the instrument for branch dynamics inside the ILTF (see Section 5). White areas denote other BLA municipalities, belonging to neither the ILTF nor the outer ring. The right panel shows the deforestation polygons recorded inside the Indigenous land over 2008 to 2024.

Using the ILTF as our unit of analysis, we aggregate bank-branch counts across the municipalities that make up each ILTF. This measure captures the intensity of local banking presence in the municipalities with direct access to the Indigenous land and is our main explanatory variable, from which we construct the year-on-year changes used in the baseline estimation (Section 5). The dependent variable is deforestation inside the Indigenous land, measured with satellite data: for each ILTF and year, we sum the deforestation polygon area that falls inside the Indigenous land boundary.

	N	Mean	SD	Min	Median	Max
Indigenous Land area (1000 km²)	381.0	3.0	9.3	0.0	0.4	96.6
ILTF area (1000 km²)	381.0	52.0	58.6	2.1	28.6	424.0
Indigenous Land population (2022, thousands)	358.0	1.2	3.0	0.0	0.4	27.2
ILTF population (2022, thousands)	381.0	88.7	98.9	4.1	54.2	727.4
Indigenous Land population density (2022, pop./km²)	358.0	14.5	116.8	0.0	0.8	1996.7
ILTF population density (2022, pop./km²)	381.0	3.6	5.9	0.2	1.6	72.4
Number of municipalities per ILTF	381.0	2.3	1.7	1.0	2.0	14.0
Average annual IL deforestation (km²)	381.0	0.7	2.3	0.0	0.1	28.6
Accumulated IL deforestation (2008–2024, km²)	381.0	11.2	39.7	0.0	1.3	485.8
Average branches per municipality (2022)	381.0	2.4	3.1	0.0	1.9	34.1
Total branches per ILTF (2022)	381.0	5.3	6.6	0.0	3.0	48.0

Table 2: Descriptive statistics for Indigenous Land Territorial Footprints (ILTFs) and associated Indigenous lands. Areas and populations are reported in thousands. Population variables use the 2022 IBGE census and therefore have smaller observations than the full ILTF sample.

Table 2 reports descriptive statistics for the Indigenous lands in our sample and their ILTFs. Indigenous lands average about 3,041 km^2 but vary widely in size, with the largest comparable in area to Portugal. The Brazilian Amazon is sparsely populated overall, reflected in the low average population density across ILTFs; within this low-density setting, average density is higher inside the lands than in their surrounding footprints. The 2022 Indigenous Population Census (Instituto Brasileiro de Geografia e Estatística, 2022) provides population measures for 358 of the 381 lands in our sample, among which average density is 14.5

pop/km^2 inside the Indigenous land, compared with 3.4 in the corresponding ILTF. Bank branch coverage is likewise thin: in 2022 the average municipality had 2.4 branches and the average ILTF 5.3, yet some ILTFs have no branches at all. Deforestation inside Indigenous lands is also highly heterogeneous, averaging 0.7 km^2 of forest loss per year and 11.2 km^2 cumulatively between 2008 and 2024.

The data used in this paper originate from a variety of Brazilian institutions. Deforestation figures come from the Project for Monitoring Deforestation in the Legal Amazon by Satellite (PRODES) (Instituto Nacional de Pesquisas Espaciais (INPE), 2024), conducted by the National Institute for Space Research of Brazil (INPE). The PRODES dataset provides annual, wall-to-wall maps of clear-cut deforestation across the BLA, produced through visual interpretation of high-resolution satellite imagery from the Landsat series. Each detected deforestation event is represented as a geo-referenced polygon, with a minimum mapping unit of 0.065 km^2. Shapefiles delineating the municipal boundaries within the BLA come from the cartographic database of the Brazilian Institute of Geography and Statistics (IBGE). Spatial layers identifying officially recognized Indigenous lands and coordinates of FUNAI field offices come from the Geoprocessing and Mapping Division of FUNAI.

Information on bank branches comes from the Banco Central do Brasil and is based on mandatory monthly reports submitted by financial institutions at the municipality level. We aggregate these monthly observations into an annual average stock of branches for each municipality, smoothing transient openings and closures within a given year. The resulting dataset provides a municipality-level time series of average bank-branch stocks for the Amazon region covering 2008 to 2024.

Population counts and municipal land area are obtained from IBGE.

Table A1.1 in Appendix A reports summary statistics for the main variables used in the analysis, and Table A1.2 summarizes all variable definitions and data sources.

5 Empirical strategy and results

This section documents the relationship between local banking presence and deforestation inside Indigenous lands, with a focus on the estimation of the channels through which it runs. Because branches sit outside the lands, any effect must operate indirectly, through the mediators of the model in Section 3.1. For each ILTF and year, we measure bank presence as the total number of branches across all municipalities that make up the footprint, denoted $B_{i,t}$. Our main regressor, $\Delta B_{i,t}$, is the year-on-year change in this branch count.

5.1 Empirical strategy

Because effects may unfold gradually, we estimate local-projection regressions (Jordà, 2005; Jordà and Taylor, 2025) of cumulative deforestation up to horizon $H = 5$ on the instrumented contemporaneous change in

bank branches. Each specification includes ILTF and year fixed effects and controls for lagged deforestation inside Indigenous lands and in non-protected areas. Formally, for each $H \in \{1, \ldots, 5\}$ we estimate

$$\sum_{h=1}^{H} D_{i,t+h} = \alpha_t + \alpha_i + \beta_H \, \Delta \hat{B}_{i,t} + \boldsymbol{\gamma}_H' \, \mathbf{X}_{i,t} + \nu_{i,t}^{(H)}, \qquad H \in \{1, \ldots, 5\}. \tag{2}$$

where $D_{i,t}$ denotes the annual clear-cut area measured strictly inside the Indigenous land boundary associated with ILTF i. The dependent variable $\sum_{h=1}^{H} D_{i,t+h}$ aggregates deforestation over the subsequent H years, so β_H is interpreted as the change in cumulative Indigenous land deforestation up to horizon H following a one-unit increase in the number of local branches in year t. The regressor $\Delta \hat{B}_{i,t}$ denotes the instrumented yearly change in ILTF branch stock (the instrumental-variable strategy is described below). The control vector $\mathbf{X}_{i,t}$ includes lagged deforestation inside Indigenous lands and in surrounding non-protected areas, capturing persistence and common local deforestation dynamics. The fixed effects α_i and α_t absorb what is constant within an ILTF over the sample and what is common across ILTFs in a given year. Appendix B maps them onto the structural variables they stand in for, and states which residual variation they leave behind.

To address endogeneity in branch location and dynamics, we instrument $\Delta B_{i,t}$ with the contemporaneous change in bank branches in the outer ring of ILTF i, defined as the set of municipalities that border the ILTF but do not intersect the Indigenous land itself. The identifying assumption is that outer-ring branch changes affect deforestation inside the Indigenous land only through their effect on branch presence within the ILTF. This exclusion restriction is plausible because outer-ring branch openings and closures mainly reflect network-wide expansion or contraction driven by institutional factors and regional trends, rather than conditions inside the protected territory or its footprint. The instrument cannot affect the outcome through any mechanical link: deforestation is measured strictly inside the Indigenous land boundary, and outer-ring branches lie outside it by construction.

A remaining concern is a common regional shock correlated with both outer-ring and ILTF branches. The year fixed effects and lagged deforestation controls absorb any such common shock, and Section 6.2 probes this directly by saturating the specification with state-by-year and mesoregion-by-year fixed effects. First-stage statistics reported in Table 3 show a cluster-robust Wald statistic of at least 38 at every horizon, far above conventional weak-instrument thresholds.

5.2 Total effect of bank branches on Indigenous land deforestation

Table 3 also presents the 2SLS estimates for horizons from one to five years. The coefficients on $\Delta \hat{B}_{i,t}$ are negative and statistically significant at conventional levels for all horizons. At $H = 1$, an additional bank branch lowers cumulative deforestation inside Indigenous lands by about 0.71 km^2 over the subsequent year.

The effect builds with the horizon, reaching 2.7 km^2 at $H = 3$ and staying flat through $H = 5$. This pattern indicates that the effect of local bank presence on deforestation is persistent rather than transitory: it is realized within about three years of a branch opening and holds through the five-year window we estimate. The lagged deforestation controls show that deforestation inside Indigenous lands is somewhat persistent over time, while lagged deforestation in nearby non-protected areas is generally unrelated to current Indigenous land deforestation, indicating that deforestation dynamics inside Indigenous territories are distinct from those in the surrounding forest.

Overall, the baseline estimates imply that changes in the local stock of bank branches are an economically meaningful determinant of forest loss in Indigenous lands.[7] To gauge magnitudes, we benchmark the estimates against baseline deforestation levels. The five-year estimate of roughly 2.6 km^2 implies a reduction equivalent to about 4 years of the average land's annual deforestation, so that even a single branch opening constitutes a meaningful shift in cumulative forest loss over the medium run. Deforestation inside Indigenous lands is highly skewed, so the estimate is carried by the lands under active clearing pressure; where there is no measurable clearing there is nothing for a branch to prevent. Having established this relationship, Section 5.3 turns to the mechanisms that can rationalize it, using the structural framework of Section 3.1 to guide a mediation analysis along the relevant channels.

Dependent Variables:	1 yr	2 yr	3 yr	4 yr	5 yr
Variables					
Δ Branches	-0.711**	-1.882**	-2.742**	-2.769**	-2.644**
	(0.341)	(0.767)	(1.194)	(1.329)	(1.230)
Deforest. Indigenous land (t-1)	0.180***	0.308**	0.203	0.026	-0.289
	(0.047)	(0.154)	(0.209)	(0.165)	(0.341)
Deforest. non-protected areas (t-1)	-0.002**	-0.009*	-0.013	-0.015	-0.018
	(0.001)	(0.005)	(0.009)	(0.011)	(0.012)
Year FE	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	5,715	5,334	4,953	4,572	4,191
R^2	0.407	0.493	0.548	0.591	0.648
Wald (1st stage), Δ Branches	40.249	39.608	38.301	41.928	45.560
First stage, outer-ring branches	0.126 (0.020)	0.123 (0.020)	0.122 (0.020)	0.128 (0.020)	0.133 (0.020)

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Influence of bank presence on deforestation in Indigenous lands.

[7] These effects are a result of both federal retail and private retail banks. Further details are found in Appendix D.

5.3 Mediation strategy

We decompose the total effect of bank presence on deforestation into the part that is mediated by bank-induced changes in social capital and the part that is mediated through everything else. Following Heckman and Pinto (2024), the causal effects of interest are defined by fixing variables in a hypothetical model, and identification maps them to the data; the formal statement is in Appendix B. Writing B_i for bank presence, S_i for social capital and D_i for deforestation, and suppressing fixed effects and controls, the linear model is

$$\begin{aligned} \Delta B_i &= \pi\, \Delta B_j + V_B + \epsilon_B, \\ S_i &= \beta_S^B\, \Delta B_i + \lambda_B V_B + \lambda_D V_D + \epsilon_S, \\ D_i &= \tau\, \Delta B_i + \beta_D^S\, S_i + V_D + \epsilon_D, \end{aligned} \tag{3}$$

where V_B collects the unobserved drivers of branch placement and V_D the unobserved drivers of deforestation. The coefficient β_S^B is the effect of an additional branch on social capital, β_D^S the effect of social capital on deforestation holding branch presence fixed, and τ the effect of branches holding social capital fixed. Branches cannot clear forest or prevent its clearing themselves, so τ is not a direct effect in the usual sense: it captures the mediation that does not run through social capital, which we also call the mediation outside social capital. Proposition 5.1 below makes this precise. The total effect estimated in Section 5 decomposes as

$$\beta_D^B \;=\; \tau \;+\; \beta_S^B\, \beta_D^S. \tag{4}$$

A single instrument identifies β_D^B and β_S^B but not, on its own, the split between τ and β_D^S, because both S_i and ΔB_i are endogenous in the deforestation equation. The standard solution is a separate instrument for the mediator (Frölich and Huber, 2017), which our setting does not offer. We instead follow the single-instrument approach of Dippel et al. (2021), which rests on three assumptions; Appendix B states them formally.

- **(A1) Instrument exclusion.** The outer-ring branch change ΔB_j affects D_i only through ILTF branch presence. This is the exclusion restriction of Section 5, probed in Section 6.
- **(A2) Mediated confounding.** $V_B \perp\!\!\!\perp V_D$: the unobserved factors that drive branch placement may also move social capital, but they do not drive deforestation directly. Bank network strategy, federal branching mandates and the digitization wave plausibly shape branch counts and local civic life without acting on clearing inside the Indigenous land, while criminal-organization activity and local norms shape clearing and civic life without driving branch placement. The assumption would fail if a local shock, such as an illegal gold boom, attracted branches and changed clearing at the same time. It would also fail if the mediation outside social capital were itself driven by V_B. It is not directly testable.

- **(A3) Linearity.** The mediator and outcome equations are linear around the equilibrium of the model in Section 3.1.

Mediated confounding delivers the extra source of exogenous variation. Since the endogeneity of ΔB_i runs through V_B, which does not enter the deforestation equation, ΔB_i is exogenous there and can be included as a control. Conditional on ΔB_i, ΔB_j remains a valid instrument for S_i, and it remains relevant as long as the confounder V_B moves social capital ($\lambda_B \neq 0$): holding branch presence fixed, a higher outer-ring change implies a smaller confounder component, which shifts S_i. This requirement is plausible in our setting. Federal mandates directed new branches to underbanked towns, which also lacked civic infrastructure, and the closures that followed the spread of digital banking concentrated in towns with shrinking populations, where civic life was thinning. We therefore estimate three regressions on a common sample. The first is the baseline specification, Equation 2, which identifies β_D^B. The second and third are

$$S_{i,t} \;=\; \beta_S^B \,\widehat{\Delta B}_{i,t} \;+\; \alpha_i \;+\; \alpha_t \;+\; \boldsymbol{\gamma}'\mathbf{X}_{i,t} \;+\; u_{i,t}^S, \tag{5}$$

$$\sum_{h=1}^{H} D_{i,t+h} \;=\; \beta_D^S \,\widehat{S}_{i,t} \;+\; \tau\,\Delta B_{i,t} \;+\; \alpha_i \;+\; \alpha_t \;+\; \boldsymbol{\gamma}_H'\mathbf{X}_{i,t} \;+\; u_{i,t}^D, \qquad H \in \{1,\ldots,5\}, \tag{6}$$

where $\Delta B_{j,t}$ instruments $\Delta B_{i,t}$ in Equation 5 and $S_{i,t}$ in Equation 6. Branch presence $\Delta B_{i,t}$ also appears in Equation 6 as a control, and the fixed effects and controls are those of Section 5. We estimate Equation 5 and Equation 6 jointly by GMM, clustering at the Indigenous-land level, so that the indirect effect $\beta_S^B\beta_D^S$ has a delta-method standard error that uses the full cross-equation covariance. On a common sample, Equation 6 implies that the estimated total effect equals $\hat{\tau} + \hat{\beta}_S^B\hat{\beta}_D^S$ exactly, so the content of the exercise is how the total splits.

Keeping $\Delta B_{i,t}$ in Equation 6 is what makes the split informative. The control does not stand for a direct effect of branches on the forest; it absorbs the mediation that does not run through social capital. Without it, and writing $Z \equiv \Delta B_{j,t}$ for the instrument, the IV estimate $\hat{\beta}_D^S$ would equal $\mathrm{Cov}(Z,D)/\mathrm{Cov}(Z,S)$ and the product with $\hat{\beta}_S^B = \mathrm{Cov}(Z,S)/\mathrm{Cov}(Z,\Delta B)$ would reduce to $\mathrm{Cov}(Z,D)/\mathrm{Cov}(Z,\Delta B)$, the total effect, for any mediator the instrument moves. With the control, β_D^S is identified only from the variation in social capital that branch presence does not explain, and $\tau = 0$ becomes a testable restriction rather than an assumption. We report the cluster-robust first-stage Wald statistic for both instrumented legs. As in any linear IV design with heterogeneous effects, the estimates describe ILTFs whose branch count responds to outer-ring network dynamics.

Proposition 5.1 states what the coefficients of Equation 6 measure when branches can also act through mediators other than social capital. It is derived from the single-instrument mediation result of Dippel et al. (2021), who show that, under mediated confounding, the instrument identifies the effect of the mediator once

the treatment enters the outcome equation as a control. Following Heckman and Pinto (2015) and Heckman and Pinto (2024), the effects in the proposition are defined by fixing variables in the hypothetical model of Appendix B. Our addition is to allow for an unknown number of further mediators and to characterize what the coefficient on the treatment then measures.

Proposition 5.1 (Coefficient on branch presence in the second leg, following Dippel et al. (2021))**.** *Let $Z \equiv \Delta B_{j,t}$, and let all variables be net of the fixed effects and controls of Equation 6. Suppose that deforestation and social capital depend on m mediators $M_1, \ldots, M_m$ that cause social capital and n mediators $N_1, \ldots, N_n$ that respond to it:*

$$D = \gamma\,\Delta B + \delta\,S + \sum_{k=1}^{m} \theta_k M_k + \sum_{j=1}^{n} \phi_j N_j + U_D, \qquad S = \psi\,\Delta B + \sum_{k=1}^{m} \pi_k M_k + U_S,$$

$$M_k = \alpha_k\,\Delta B + U_k, \qquad N_j = a_j\,\Delta B + \rho_j\,S + V_j,$$

so that mediators may cause social capital or respond to it, although two-way loops are not considered in this example, and γ is the direct effect of branches on deforestation. Suppose also that (i) Z is uncorrelated with U_D, U_S, each U_k and each V_j; (ii) ΔB is uncorrelated with $U_D + \sum_k \theta_k U_k + \sum_j \phi_j V_j$; and (iii) $\mathrm{Cov}(Z, \Delta B) \neq 0$ and $\mathrm{Cov}(Z, S)\,\mathrm{Var}(\Delta B) - \mathrm{Cov}(Z, \Delta B)\,\mathrm{Cov}(\Delta B, S) \neq 0$. Then the IV estimands of Equation 5 and Equation 6 are

$$\beta_S^B = \psi + \sum_{k=1}^{m} \pi_k \alpha_k, \qquad \beta_D^S = \delta + \sum_{j=1}^{n} \phi_j \rho_j, \qquad \tau = \gamma + \sum_{k=1}^{m} \theta_k \alpha_k + \sum_{j=1}^{n} \phi_j a_j,$$

and the total effect satisfies $\beta_D^B = \beta_S^B \beta_D^S + \tau$. In particular, if branches have no direct effect on deforestation ($\gamma = 0$), the coefficient on $\Delta B_{i,t}$ in Equation 6 is $\tau = \sum_k \theta_k \alpha_k + \sum_j \phi_j a_j$, which collects every path from branches to deforestation that does not pass through social capital. The product $\beta_S^B \beta_D^S$ collects every path that does, the mediation through social capital.

The proof is in Appendix B. Conditions (i) and (ii) are (A1) and (A2) written for the model with further mediators; (ii) asks that the unobserved drivers of those mediators be unrelated to branch placement, as noted under (A2). Condition (iii) is relevance. Its first part is the usual first stage. The determinant in its second part is non-zero exactly when the outer-ring change still moves social capital after branch presence is partialled out, which is the requirement that the confounder move social capital ($\lambda_B \neq 0$).

The proposition separates what identification requires from what interpretation requires. The coefficient on social capital is identified whether or not branches have a direct effect on the forest. Reading τ as the mediation outside social capital requires $\gamma = 0$. That premise is substantive but plausible: branches sit in town centers outside the land boundary, so any effect on clearing inside the land must travel through the

people and resources that branches change. If it failed, τ would add a direct effect to the mediation outside social capital, while the estimate of the mediation through social capital would be unaffected.

We do not need to know what the further mediators are, or how many there are. A mediator that causes social capital, such as the venues that local credit finances, contributes to the mediation through social capital along the paths that run through it, and to τ along the paths that do not.

5.4 Social capital: results

Section 5.2 established that bank presence lowers deforestation. We now show that social capital carries that effect. The mediator is the index of Section 3.2, the first principal component of the valid vote share and non-profit association employment. Figure 7 decomposes the total effect of an additional branch into the part that runs through the index and the part that does not, at each horizon.

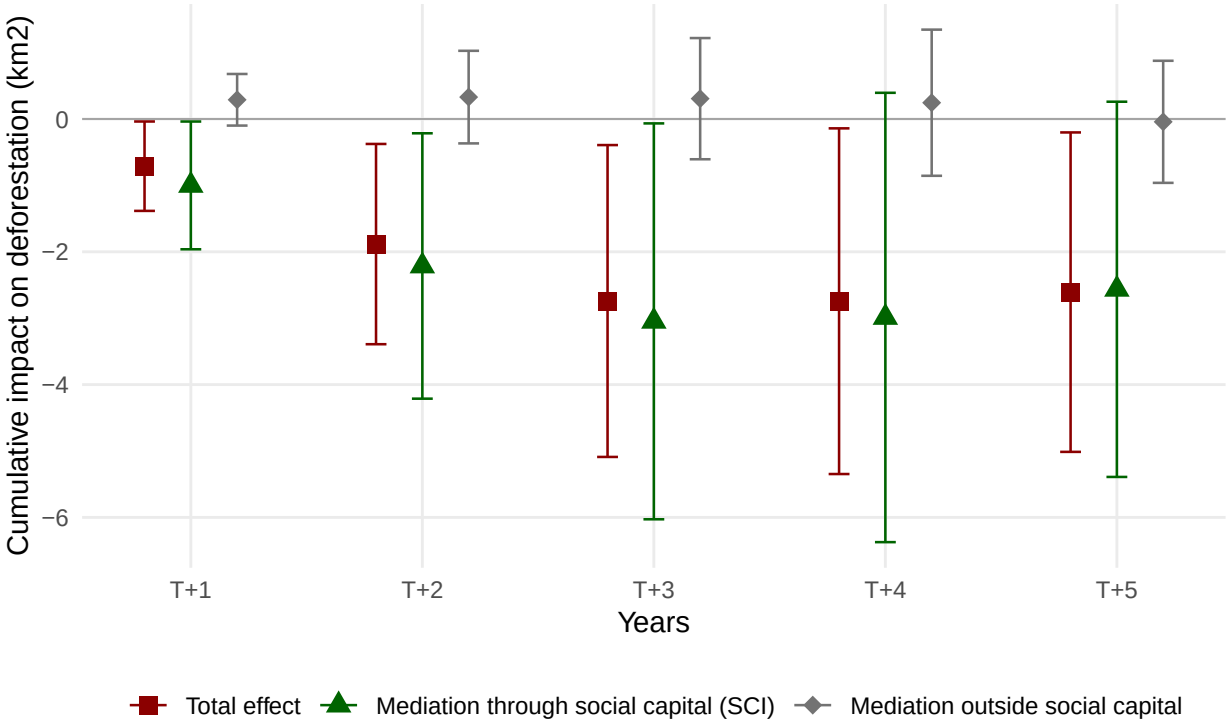


Figure 7: Cumulative deforestation inside Indigenous lands in response to one additional bank branch, decomposed into the mediation through social capital and the mediation that does not run through it. The mediator is the social capital index. All specifications include year and ILTF fixed effects and lagged deforestation controls, with standard errors clustered at the Indigenous-land level. The channels are estimated by joint GMM with delta-method intervals, on the same sample as the total, so the two sum to it exactly.

The indirect effect through social capital tracks the total effect at every horizon. At three years, an additional branch lowers cumulative deforestation by 2.74 km^2 in total, and the social capital channel

accounts for 3.05 km^2 (standard error 1.52). The mediation outside social capital ranges from -0.04 to 0.33 km^2 across horizons and is never statistically different from zero. The point estimates of the channel slightly exceed the total at short horizons, which leaves room for mediation outside social capital that works in the opposite direction. This pattern is what the model in Section 3.1 suggests: a stronger civic fabric raises the reservation utility of potential perpetrators and depresses participation in illegal forest clearing.

The two legs behind the decomposition are estimated equation by equation in Table A3.1 and jointly by GMM in Table A3.2, both in Appendix C. They instrument different regressors, so each has its own first stage: the cluster-robust Wald statistic is 35 for branch presence in the first leg, and ranges from 13 to 42 for social capital in the second. That appendix also reports Anderson-Rubin confidence sets, which remain valid when instruments are weak. Social capital still carries the effect when the index is recomputed on a common scale across lands (Appendix C), and when regional shocks are absorbed with the mesoregion-by-year fixed effects of Section 6.2 (Table A3.3).

The decomposition rests on mediated confounding, which the data cannot test. We ask instead how large a violation would have to be to overturn the result. Suppose the branch change is correlated with the unobserved determinants of deforestation, so that the second moment condition of Equation 6 fails by κ km^2 per branch. The total effect is unchanged, and the indirect effect through social capital moves linearly in κ. At three years, the indirect effect falls to zero only if κ reaches 2.38 km^2 per branch, 87 percent of the total effect in absolute value. The confounding would have to be almost as large as the entire effect it is meant to explain away. Appendix C reports this threshold at every horizon.

5.5 How bank branches foster social capital

Two complementary mechanisms link a bank branch to local social capital. First, in sparsely populated regions such as the Amazon, a bank branch serves as an attractor to people who live outside the main city center, putting them in contact with other residents locally. Second, local banks more easily finance the small enterprises that make routine social interaction possible: restaurants, bars, entertainment venues, and supermarkets. In either case, the branch is the catalyst for what Putnam et al. (1993) calls the "horizontal" interactions through which trust and norms of cooperation form. We can test only the second: the first would require data on who travels to a branch, which we do not have.

Table 4 tests the second mechanism. Each column reports the instrumented effect of an additional bank branch on the log stock per 100,000 inhabitants of one category of business. The response is concentrated in restaurants and bars and in supermarkets, while entertainment venues, artistic activities and hair salons do not move. The two that respond are the ones with the most to finance, such as inventory and payroll. Bank branches raise the supply of some everyday meeting places, the kind of small business through which a community accumulates social capital.

Dependent Variables:	Rest. & bars	Supermarkets	Hair salons	Entertainment	Artistic activities
Variables					
Δ Branches	0.104***	0.073**	-0.014	0.019	-0.022
	(0.040)	(0.029)	(0.057)	(0.043)	(0.039)
Year FE	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes
Lagged deforestation	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	5,543	6,096	3,674	2,896	2,992
R^2	0.904	0.801	0.834	0.760	0.641
Wald (1st stage), Δ Branches	38.067	35.822	31.037	32.746	26.786

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: Impact of local bank presence on selected local businesses. The dependent variable is the log of each category's stock per 100,000 inhabitants.

This is what the social-capital channel looks like up close: not activity in general, but the kind people take part in routinely. Carrying the venues forward as mediators in their own right reproduces the deforestation result: Table A5.1 in the Appendix E instruments each category in turn, and the two that branch entry actually moves, restaurants and bars and supermarkets, lower cumulative deforestation inside Indigenous lands. The categories that branch entry leaves flat move the mediator too little for that second step to speak to deforestation either way.

6 Robustness

Our key identifying assumption is that outer-ring branch changes shift ILTF branch presence but do not otherwise affect deforestation inside the Indigenous land except through those ILTF branches. One might worry, however, that even with the long distances between municipalities, outer-ring branch changes could be correlated with broader regional developments that affect Indigenous-land deforestation directly. Changes in state-level capacity or willingness to enforce Indigenous rights, or to counter criminal organizations operating in the forest, could affect both regional bank presence (including the outer ring) and deforestation inside Indigenous lands. This section presents results that reinforce the credibility of the identifying assumption.

We assess the baseline result of Section 5 in six ways. To probe the exclusion restriction, we show that instrumented branch changes do not predict past deforestation, add state-by-year and mesoregion-by-year fixed effects that absorb shocks varying across regions within a year, and rebuild the instrument from farther rings of municipalities. Beyond identification, we restrict the sample by legal-protection status, since

recognition is still in process for some lands. We also allow for footprints and outer rings that overlap in the inference, and split the effect by federal enforcement presence, which may substitute for the social sanctions that carry the channel.

6.1 Falsification with past trends

Our first falsification test asks whether current instrumented branch changes predict past deforestation inside Indigenous lands. Under the exclusion restriction they should not: if the instrument were merely tracking confounders that drive forest loss, it would correlate with deforestation that predates the branch change. Table 5 reports the estimates. None of the three coefficients is significant at the 5 percent level. The coefficient on deforestation two years earlier is not negligible relative to the one-year effect, so we read this result as supporting although not quite conclusive evidence in favor of the exclusion restriction; the remainder of this section presents sharper tests.

Dependent Variables:	Defor. (t-2)	Defor. (t-3)	Defor. (t-2, t-3)
Variables			
Δ Branches	-0.434	-0.020	-0.382
	(0.271)	(0.292)	(0.515)
Deforest. Indigenous land (lagged)	0.367***	0.234***	0.643***
	(0.071)	(0.079)	(0.138)
Deforest. non-protected areas (lagged)	0.004*	0.003	0.008**
	(0.002)	(0.002)	(0.004)
Year FE	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes
Fit statistics			
Observations	5,715	5,334	5,334
R^2	0.480	0.413	0.548
Wald (1st stage), Δ Branches	35.336	35.149	35.149

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 5: Falsification analysis with the effect of instrumented branch changes on past deforestation inside Indigenous lands.

6.2 Regional fixed-effect saturation

The baseline includes year and ILTF fixed effects, which absorb common time shocks and time-invariant ILTF characteristics. This configuration, however, does not fully absorb shocks that vary across regions within a year: federal credit programs, commodity-price cycles, or development initiatives that typically operate

at the level of states or mesoregions[8]. Other regional dynamics not captured by the baseline could play a similar role: local state capacity and coordination with environmental enforcement, political organization, land-market pressure, criminal activity, and the density of transport and services around Indigenous-land borders. If such shocks are correlated with both outer-ring branch dynamics and deforestation inside Indigenous lands, the baseline coefficient would be biased.

To address this concern, we augment the local projection estimates with state-by-year and mesoregion-by-year fixed effects, identifying the bank effect from within-cell variation only.[9]

	Cumulative deforestation inside Indigenous lands				
	1 yr	2 yr	3 yr	4 yr	5 yr
Coefficient on Δ Branches (instrumented), by fixed-effect specification					
Year + ILTF FE	-0.71**	-1.88**	-2.74**	-2.77**	-2.64**
	(0.34)	(0.77)	(1.19)	(1.33)	(1.23)
State-year + ILTF FE	-0.70*	-1.87**	-2.70**	-2.72*	-2.79*
	(0.39)	(0.86)	(1.37)	(1.52)	(1.45)
Mesoregion-year + ILTF FE	-1.12**	-2.52***	-3.51**	-3.76**	-4.23**
	(0.49)	(0.97)	(1.44)	(1.67)	(1.76)
Lagged deforestation controls	Yes	Yes	Yes	Yes	Yes
Wald (1st stage)					
Year + ILTF FE	40.2	39.6	38.3	41.9	45.6
State-year + ILTF FE	31.9	31.1	30.4	33.3	35.7
Mesoregion-year + ILTF FE	24.5	23.7	23.1	24.8	25.1
R^2	0.41–0.46	0.49–0.54	0.55–0.59	0.59–0.63	0.65–0.68
Observations†	5,715	5,334	4,953	4,572	4,191

Table 6: Baseline local projection estimates under regional fixed-effect saturation. Clustered (ILTF) standard errors in parentheses. R-squared is reported as a range across the three specifications. † First two specifications. Mesoregion-by-year drops Waiãpi, the only land in its mesoregion. Significance: *** 0.01, ** 0.05, * 0.1.

Table 6 reports the baseline local projection estimates under both saturations, alongside the headline specification. Point estimates are similar under state-by-year fixed effects and larger under mesoregion-by-year fixed effects: at the four-year horizon, the coefficient is -2.77 in the headline specification (Table 3), -2.72 under state-by-year and -3.76 under mesoregion-by-year, about 36% larger in magnitude than the headline. The cluster-robust first-stage Wald statistic falls as more regional variation is absorbed: from at least 38 across horizons in the headline specification to between 30 and 36 under state-by-year and between 23 and 26 under mesoregion-by-year fixed effects, remaining above conventional weak-instrument thresholds throughout. The larger estimate under mesoregion-by-year fixed effects is consistent with regional shocks

[8] the IBGE administrative subdivision between municipality and state

[9] When an Indigenous land's territory spans more than one state or mesoregion, we assign it to the state and mesoregion containing the largest share of its area.

that correlate with both branch dynamics and deforestation and attenuate the baseline coefficient toward zero. Under regional saturation the baseline estimates are, if anything, conservative. The mediation survives the same saturation: under mesoregion-by-year fixed effects the effect through social capital is larger and significant at every horizon, while the mediation outside social capital stays indistinguishable from zero (Table A3.3).

6.3 Sample restriction by legal-protection status

The baseline sample includes all 381 Indigenous lands in the BLA regardless of administrative phase. By the end of 2007, the year before the panel begins, 87% are already legally protected[10] against incursion; of the remaining 50, 38 become protected by 2024. A natural concern is that our estimates are driven by lands whose administrative recognition is still in process, where the protection mechanism we propose cannot operate.

To address this concern, we restrict the sample to the 336 ILTFs whose Indigenous land was legally protected in every year of the panel, 2008 to 2024, dropping those not continuously protected. Table 7 reports the baseline local projection estimates for this subsample. Estimates are nearly identical to the headline. At the four-year horizon, the coefficient is -2.97 against -2.77 in the full sample, confirming that the result is not driven by lands whose administrative recognition is still in process. First-stage Wald statistics remain comfortably above conventional thresholds.

Dependent Variables:	1 yr	2 yr	3 yr	4 yr	5 yr
Variables					
Δ Branches	-0.697*	-1.834**	-2.760**	-2.970**	-2.726**
	(0.355)	(0.785)	(1.270)	(1.478)	(1.384)
Year FE	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes
Lagged deforestation controls	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	5,040	4,704	4,368	4,032	3,696
R^2	0.456	0.542	0.585	0.616	0.666
Wald (1st stage), Δ Branches	34.242	33.885	33.252	35.901	38.940

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 7: Baseline local projection estimates restricted to ILs continuously legally protected during 2008-2024

[10] An Indigenous land is *legally protected* if it has reached at least the *Declarada* phase by that year, or any of its parallel-track equivalents (*Reserva Indígena*, *Restrição de Uso*, *Dominial Indígena*). See Appendix A for the full classification and the data construction.

6.4 Alternative instruments from farther rings

The outer ring borders the ILTF, and its towns may share part of the local economy that reaches the forest: perpetrators may bank, finance machinery or sell illegally mined gold there. If so, outer-ring branches would affect deforestation directly, and the exclusion restriction would be violated. We build two instruments from municipalities that are less exposed to this concern. The first is the branch change in the second ring of municipalities, the neighbors of the outer ring that are neither in the ILTF nor in its outer ring. The second keeps only the outer-ring municipalities whose centroid lies more than 127 km from the Indigenous land, the median distance across pairs. Following Pollmann (2026), the comparison asks whether the estimate depends on which nearby locations supply the variation.

	Cumulative deforestation inside Indigenous lands				
	1 yr	2 yr	3 yr	4 yr	5 yr
Panel A: Outer ring (baseline)					
Δ Branches	-0.711^{**}	-1.882^{**}	-2.742^{**}	-2.769^{**}	-2.644^{**}
	(0.341)	(0.767)	(1.194)	(1.329)	(1.230)
Wald (1st stage)	40.2	39.6	38.3	41.9	45.6
Anderson-Rubin 95% set	$[-1.44, -0.05]$	$[-3.53, -0.39]$	$[-5.31, -0.41]$	$[-5.56, -0.18]$	$[-5.17, -0.18]$
Panel B: Second ring					
Δ Branches	-4.318^{**}	-10.727^{*}	-15.947	-14.423^{*}	-12.132^{*}
	(2.177)	(6.156)	(10.164)	(8.362)	(6.635)
Wald (1st stage)	6.1	5.0	3.9	8.8	16.3
Anderson-Rubin 95% set	$[-20.32, -1.38]$	$(-\infty, -2.42]$	$(-\infty, -2.73]$	$[-48.29, -0.63]$	$[-30.38, -0.19]$
Panel C: Far outer-ring municipalities					
Δ Branches	-2.197^{*}	-6.421^{*}	-10.834^{*}	-10.594^{*}	-9.560^{*}
	(1.178)	(3.482)	(6.467)	(6.284)	(5.286)
Wald (1st stage)	9.1	8.0	6.5	8.1	12.1
Anderson-Rubin 95% set	$[-6.61, -0.14]$	$[-21.22, -0.50]$	$[-46.73, -0.16]$	$[-36.04, 0.40]$	$[-24.89, 0.48]$
Observations	5,715	5,334	4,953	4,572	4,191

ILTF and year fixed effects and lagged deforestation controls in every panel
Standard errors clustered by Indigenous land
*Signif. codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Alternative instruments for the effect of one additional ILTF branch on cumulative deforestation inside the Indigenous land. Each panel instruments the branch change with a different set of municipalities.

Table 8 reports the estimates. Moving the instrument farther from the land does not change the picture: both farther rings give negative effects, and their Anderson-Rubin sets exclude zero at all five horizons for the second ring and at 3 of the five horizons for the far outer-ring municipalities. The first stages there are weak, with Wald statistics between 3.9 and 16.3, so we rely on those sets rather than on conventional intervals, which would not have correct coverage.

The sets are much wider than the baseline's, so they bound the sign of the effect rather than its size. Branch changes farther away move ILTF branches only in a few lands, so these estimates describe a different and smaller set of lands.[11] Appendix F reports a third instrument that does not use the geography of the outer ring: the expansion of each bank's network elsewhere in the BLA, weighted by its presence in the ILTF before the sample (Borusyak et al., 2022). It draws on an entirely different source of variation and gives -3.60 km^2 at three years against -2.74 km^2 in the baseline.

6.5 Inference

ILTFs and their outer rings overlap: a municipality can belong to the footprint of one land and the outer ring of another. Clustering by Indigenous land may then understate the dependence across observations. Table 9 reports two alternatives, which differ in kind. The first widens the cluster. Mesoregions nest most overlapping footprints, and clustering there gives standard errors between 0.29 and 1.26, below the baseline at every horizon. The sample has only 25 mesoregions, and the cluster-robust estimator is biased downward when the number of clusters is small.

The second drops the variance model altogether. Randomization inference reassigns each land's outer-ring branch series to another land in the same state, keeping the series intact, and compares the reduced-form coefficient with the distribution of coefficients from 500 such reassignments. The p-values lie between 0.018 and 0.048. The test needs no assumption about the covariance of the errors, only that the series are exchangeable across lands within a state under the null of no effect, and it therefore accommodates the overlap directly.

	Cumulative deforestation inside Indigenous lands				
	1 yr	2 yr	3 yr	4 yr	5 yr
Panel A: Second-stage estimate and cluster-robust standard errors					
Δ Branches (instrumented)	−0.711	−1.882	−2.742	−2.769	−2.644
Standard error, clustered by Indigenous land	(0.341)	(0.767)	(1.194)	(1.329)	(1.230)
Standard error, clustered by mesoregion	(0.295)	(0.716)	(1.106)	(1.259)	(1.193)
Panel B: Reduced form and randomization inference					
Outer-ring branch change	−0.090	−0.232	−0.333	−0.355	−0.353
p-value, 500 reassignments	0.022	0.018	0.018	0.034	0.048
Randomization inference reassigns the outer-ring series across lands within the same state (500 draws, reduced form)					

Table 9: Alternative inference for the baseline outer-ring estimate of Table 8. The coefficient is shown without significance stars because the table compares inference procedures rather than estimates.

[11] An over-identification test with both rings as instruments rejects at every horizon, but the available statistic assumes homoskedastic independent errors and its null also requires the two to identify the same parameter.

6.6 Federal enforcement presence and dynamic specification

Social sanctions and formal enforcement can substitute for or complement each other. We split the sample by whether the ILTF hosted a FUNAI office in 2008. At three years, the effect is -2.98 km^2 in ILTFs with a FUNAI office and 0.04 km^2 (standard error 0.26) in those without one (Table 10, Panel A). Social capital and formal enforcement are complements in our setting rather than substitutes. A plausible reason is that communities protect the forest by reporting and supporting enforcement, which requires an agency to act on the reports (Sampson et al., 1997; Rustagi et al., 2010).

Finally, the local projections include lagged deforestation controls. Adding two lags of the instrument and a second lag of deforestation, as recommended for local projections with external instruments (Stock and Watson, 2018), gives -2.67 km^2 at three years (Panel B). Dropping the lagged deforestation controls, which avoids the bias that lagged outcomes can introduce in panels with fixed effects, gives a smaller effect of -0.88 km^2 (standard error 0.42). Deforestation is persistent, and without the lagged controls that persistence enters the error term.

	Cumulative deforestation inside Indigenous lands				
	1 yr	2 yr	3 yr	4 yr	5 yr
Panel A: Federal enforcement presence					
No FUNAI office in 2008	0.170	0.458	0.039	0.032	−0.534
	(0.215)	(0.556)	(0.262)	(0.272)	(0.539)
FUNAI office in 2008	−0.762**	−2.053**	−2.980**	−3.016**	−2.798**
	(0.363)	(0.828)	(1.315)	(1.483)	(1.364)
Panel B: Dynamic specification					
Lagged instrument and deforestation controls	−0.530	−1.727**	−2.675**	−2.760**	−3.249**
	(0.368)	(0.833)	(1.260)	(1.342)	(1.490)
No lagged deforestation controls	−0.204	−0.580**	−0.877**	−1.064**	−1.283**
	(0.147)	(0.286)	(0.417)	(0.501)	(0.583)

ILTF and year fixed effects in every panel
Standard errors clustered by Indigenous land
*Signif. codes: ***: 0.01, **: 0.05, *: 0.1*

Table 10: Heterogeneity by federal enforcement presence and dynamic specification. The branch change is instrumented by the outer-ring branch change.

7 Conclusion

This paper provides causal evidence that local banking infrastructure shapes deforestation outcomes inside Indigenous lands in the Brazilian Amazon. Using local-projection IV estimates over a 2008-2024 panel, we find that each additional bank branch in an Indigenous Land Territorial Footprint reduces cumulative

forest loss inside the protected territory by approximately 2.7 km^2 within three years, an effect that persists through at least five years.

The effect runs through local social capital. Under mediated confounding, a single instrument separates the mediation through social capital from the mediation that does not run through social capital. Branch entry raises civic engagement, proxied by the valid vote share and employment in civic associations, and civic engagement lowers forest loss when branch presence is held fixed. The mediation through social capital is as large as the total effect, and the mediation that does not run through social capital is small and statistically zero. The effect is concentrated where federal enforcement is present, which suggests that it works alongside formal enforcement rather than in its place. Two complementary micro-mechanisms are consistent with this finding: branches increase the frequency of face-to-face interactions among residents in otherwise sparsely populated municipalities, and they stimulate the creation of restaurants, bars and supermarkets, both of which strengthen the local civic fabric.

These results carry implications beyond the specific setting in the Brazilian Amazon. Beyond effects on capital distribution and on financial inclusion, an underlooked effect of local *physical* banking is the support for community life. In fact, this effect appears to be strong enough to bear on environmental outcomes, particularly in regions where formal enforcement is present but thinly spread. This dimension is worth considering in policy assessments of branch retention in remote Amazonian municipalities and in other areas where the involvement of local communities can be crucial for preserving common goods such as the environment.

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Appendix A: Data, variables, and the Indigenous-land sample

Variables and data sources

Variable	Mean	SD	2.5%	25%	Median	75%	97.5%
IL deforestation, annual (km²)	0.66	3.81	0.00	0.00	0.00	0.20	4.82
IL deforestation, cumulative over 3 years (km²)	1.87	9.76	0.00	0.00	0.13	0.64	14.70
Deforestation in non-protected areas (km²)	63.20	120.16	0.08	4.12	17.38	61.81	398.43
ILTF branch change	0.23	1.08	-0.83	0.00	0.00	0.00	2.67
Outer-ring branch change	1.16	4.62	-3.25	0.00	0.00	1.08	11.33
Valid votes (%)	93.41	4.16	83.18	91.96	94.57	96.01	97.99
Assoc. employment (per 100k)	221.84	329.43	0.00	18.37	82.78	264.02	1,182.52

Table A1.1: Summary statistics of the estimation variables, pooled across Indigenous lands and years.

Variable	Data source	Definition
Deforestation increments inside Indigenous lands	PRODES / INPE	Annual clear-cut increments clipped to land boundaries, in km²
Bank branches	Banco Central do Brasil (ESTBAN)	Monthly municipal counts, averaged within the year, differenced, summed over the ILTF
Indigenous land geometries and administrative phase	FUNAI	Land polygons and phase dates; define the ILTF and legal protection
FUNAI office locations	FUNAI	Regional coordinations and local technical units, counted per municipality
Amazon and municipality shapefiles	IBGE	Municipal boundaries and Legal Amazon perimeter; build the ILTF and outer ring
Election results	Superior Electoral Court (TSE)	Valid votes over valid plus blank plus null, by municipality and election
Formal employment establishments (RAIS)	Ministry of Labor and Employment	Active contracts and establishment counts by legal nature and activity class, per 100,000 residents
Population counts	IBGE	Municipal population, denominator of the per-100,000 variables; resident population by land in the 2022 Indigenous census, descriptive use only
Indigenous peoples ethnicities and languages	Instituto Socioambiental (ISA)	Ethnic groups and language families by land; descriptive use only

Table A1.2: Definitions and data sources for the variables used in the analysis.

Legal status of Indigenous lands

Legal classification of Indigenous lands

Indigenous lands recognized by Brazil's 1988 Constitution (Art. 231) follow a five-phase administrative process under Decreto 1.775/1996. A residual set instead follows legacy or parallel tracks under Law 6.001/1973 or, for isolated peoples, by a FUNAI use-restriction order. The phases of the modern pipeline are cumulative: each new act adds a layer of protection on top of the previous one rather than replacing it, so an Indigenous land at *Registered* carries all the safeguards introduced at *Declared* and *Ratified*.

Track	**Phase**	**Triggering act**
Modern pipeline (Decree 1.775/1996)	Under study *(em estudo)*	FUNAI administrative order opening studies.
	Delimited *(delimitada)*	FUNAI president approves the studies.
	Declared *(declarada)*	Declaratory order issued by the Minister of Justice.
	Ratified *(homologada)*	Presidential decree ratifying the boundaries.
	Registered *(regularizada)*	Registration at the land registry (CRI) and federal property authority (SPU).
Other tracks	Reserve *(reserva indígena)*	Decree creating an Indigenous Reserve (Law 6.001/1973).
	Use restriction *(restrição de uso)*	FUNAI order restricting third-party access to areas occupied by isolated peoples (Art. 7, Decree 1.775/1996).
	Community-owned *(dominial indígena)*	Title acquired by an Indigenous community under ordinary civil law.

Table A1.3: Phases and tracks of Indigenous-land legal status. See FUNAI (gov.br/funai) for the full administrative procedure.

Sample construction

We start from FUNAI's 2024 universe of recognised Indigenous lands and restrict to those located in the Legal Amazon, yielding 381 Indigenous lands at various stages of the administrative process described above. Phase dates are sourced from FUNAI's portarias master file and from Instituto Socioambiental (ISA), accessed in 2026.

Track	Phase	2008 (%)	2024 (%)
Modern pipeline (Decree 1.775/1996)	Under study	6.8	1.3
	Delimited	2.4	2.1
	Declared	7.3	6.8
	Ratified	11.8	2.1
	Registered	64.8	83.7
Other tracks	Reserve	1.0	1.3
	Use restriction	0.5	1.0
	Community-owned	0.8	0.8
Not yet in process		4.5	0.8
Total		**100.0**	**100.0**

Table A1.4: Share of the 381 sampled Indigenous lands in each phase or track of Table A1.3, at the start of the panel (1 January 2008) and at its end (31 December 2024). Status at a given date is the most advanced modern-pipeline phase reached by that date, or the relevant track for lands outside the pipeline.

Table A1.4 reports the distribution. At the start of the panel, nearly two-thirds of the sample is already at *Registered*, the final administrative phase, and 83.9% has reached *Declared* or beyond. This reflects the unusually well-established documentary basis for Indigenous occupation in the Brazilian Amazon: most communities in the region have a continuous, recorded presence that makes the constitutional standard of *tradicionalmente ocupada* comparatively straightforward to satisfy, and the modern pipeline tends to proceed without major contestation up to ratification. By 2024, *Registered* alone accounts for 83.7% of the sample. Most of the variation in protection status that the panel exploits occurs at the margin between *Declared* and the lower phases, where the legally-protected indicator switches on, rather than at the later ratification or registration steps.

Definitions of land protection

We classify an Indigenous land as *legally protected* in a given year once it has reached the *Declared* phase, or the equivalent step on any of the other tracks in Table A1.3: a decree creating a Reserve, a use restriction, or formal recognition as community-owned land. We use *Declared* as the threshold because it is the first government act that bars outsiders from using the land for economic activity.

Appendix B: Structural model, mediation and estimation details

The structural model

The model of the perpetrator's choice, stated in full. The main text keeps the payoff and the direction of each channel; everything else is here.

The probability of being caught depends on enforcement capacity, on a common unobserved confounder V, and on independent shocks:

$$p_i = \sigma^p(E_i, V, \epsilon_i^p). \tag{A2.1}$$

The reservation utility depends in part on local social capital:

$$\underline{U}_i = f^{\underline{U}}(S_i, \epsilon_i^{\underline{U}}) \tag{A2.2}$$

with other plausible drivers, such as personal moral stance or religious views, subsumed into $\epsilon_i^{\underline{U}}$. The remaining relationships close the system:

$$\begin{aligned} B_i &= f^B(B_j, V, \epsilon_i^B) \\ M_i &= f^M(B_i, V, \epsilon_i^M), \quad M \in \{S, C, K\} \\ L_i &= f^L(C_i, S_i, \epsilon_i^L) \\ D_i &= f^D(\mathbf{1}[U_i > \underline{U}_i], \epsilon_i^D) \end{aligned} \tag{A2.3}$$

The framework's sign restrictions are then

$$\frac{\partial \underline{U}_i}{\partial S_i} > 0, \quad \frac{\partial U_i}{\partial C_i} < 0, \quad \frac{\partial U_i}{\partial E_i} < 0, \quad \frac{\partial U_i}{\partial K_i} > 0,$$

which require three assumptions: $u' > 0$; that punishment is worse than the proceeds from clearing, $u(J) < u(L_i^{\alpha} K_i^{1-\alpha} - \ell_i)$; and that the opportunity wage rises in local economic conditions, $\partial y_i / \partial C_i > 0$. Only the restriction on K_i follows from the payoff equation alone.

What the fixed effects absorb

The structural model above contains several variables that do not appear in the empirical specifications of Section 5.3: the enforcement level E_i, the punishment J, the logistical cost of clearing forest ℓ_i, the opportunity wage y_i from legitimate activity, and the disutility of getting caught. We do not treat these as omitted variables; rather, we absorb them through the panel structure of the data and the fixed-effect specification. This section makes that mapping explicit.

The empirical specification includes ILTF fixed effects α_i and year fixed effects α_t. Together, they absorb two classes of variation: ILTF-level characteristics and common annual shocks. These are explained in more detail below.

First are the time-invariant ILTF characteristics. α_i absorbs every variable that is constant within an ILTF over the sample period. This includes the topography, hydrography and accessibility components of the logistical cost ℓ_i, the structural distance from law-enforcement infrastructure, the size and shape of the Indigenous land, the ethnic composition of the resident population, and the long-run mean of the punishment regime J. None of these vary at the annual frequency at which we measure deforestation, so a fixed-effect transformation removes them without loss.

As for the common annual shocks, absorbed by α_t, these correspond to every variable that is constant across ILTFs in a given year. This includes federal-level changes in environmental policy (e.g., changes in IBAMA's enforcement budget, federal-level prosecutorial priorities, changes in the legal framework for Indigenous land protection), national commodity-price cycles affecting the returns to illegal activity, and macroeconomic conditions affecting the opportunity wage y_i at the national level like the level of the policy interest rate Selic. The enforcement level E_i is absorbed here to the extent it is determined by federal policy. Finally, use of lagged deforestation controls absorbs persistence in local clearing pressure, including gradual shifts in the local opportunity wage y_i and slow-moving changes in the local stock of legitimate businesses.

The remaining elements of the model that are not absorbed are residual variations in these structural variables that are simultaneously (a) ILTF-specific and (b) time-varying within the sample. Three sources of such variation deserve discussion.

First, a part of enforcement dynamics may vary over time in each ILTF, consistent with the notation $E_{i,t}$. For example, the joint capacity of FUNAI offices may vary over time and with their capacity to work with each specific Indigenous communities inside the ILTF. Cross-ILTF variation in the long-run level of E_i is absorbed by ILTF fixed effects; national-level shocks to enforcement are absorbed by year fixed effects. What remains is *within-ILTF, time-varying* enforcement variation. Two arguments justify treating this as orthogonal to the instrument ΔB_j. First, enforcement decisions in the Brazilian Amazon are made by federal and state-level agencies whose budget and deployment respond to national priorities, not to bank-network dynamics in a specific outer ring. Second, Section 6.2 shows that adding state-by-year fixed effects leaves the bank-presence coefficient unchanged and that adding mesoregion-by-year fixed effects makes it larger, so any residual correlation between ΔB_j and time-varying regional enforcement would bias the baseline estimate toward zero rather than away from it.

Time-varying logistical costs present another empirical challenge. Improvements in transport infrastructure (new roads, navigable waterways) lower the cost ℓ_i of reaching the Indigenous land boundary and could, in principle, correlate with bank-branch openings (since both follow regional development). The same

fixed-effect saturation argument applies: state-by-year and mesoregion-by-year fixed effects absorb regional infrastructure cycles, and the coefficient does not weaken under that saturation. A different case is the time-varying opportunity wage. The opportunity wage y_i from legitimate activity varies across ILTFs and over time and responds to bank presence. We do not absorb y_i via fixed effects because doing so would block one of the routes through which branches act. Its response to branches is part of the mediation that does not run through social capital, τ in Section 5.3.

Under that model and Assumptions A1 to A3, the within-transformation induced by ILTF and year fixed effects renders the unobservables $\{E_i, \ell_i, J_i\}$ either constant (within ILTF or within year) or orthogonal to the instrument ΔB_j.

From the structural model to the empirical model

We follow the framework of Heckman and Pinto (2024) and the single-instrument mediation results of Dippel et al. (2021). Heckman and Pinto distinguish between an empirical model that describes how the data is generated and a hypothetical model that formalizes the thought experiment of an external manipulation of a target variable. Causal effects are defined in the hypothetical model, while identification consists of mapping hypothetical-model quantities into empirical-model distributions.

The system above is the empirical model in this sense. Social capital is the mediator we study. Every other route from branches to deforestation is collected in a path from ΔB_i to D_i that bypasses social capital; Proposition 5.1 shows that, absent a direct effect, this path is the mediation outside social capital. Writing each variable as a function of its direct causes, and in the changes we estimate, gives the recursive form

$$\begin{aligned} \Delta B_j &= g^{\Delta B_j}(\epsilon_{B_j}) \\ \Delta B_i &= g^{\Delta B_i}(\Delta B_j,\, V_B,\, \epsilon_{B_i}) \\ S_i &= g^{S}(\Delta B_i,\, V_B,\, V_D,\, \epsilon_S) \\ D_i &= g^{D}(S_i,\, \Delta B_i,\, V_D,\, \epsilon_D), \end{aligned} \tag{A2.4}$$

where V_B and V_D are unobserved confounders that we discuss below, ΔB_j denotes the branch change in the outer-ring municipalities, and the error terms $\{\epsilon_{B_j}, \epsilon_{B_i}, \epsilon_S, \epsilon_D\}$ are mutually independent. ΔB_i enters g^D even though branches sit in the urban centers of municipalities surrounding the Indigenous land and cannot clear forest or prevent its clearing. Its role there is to carry every channel other than social capital.

To define the causal effect of social capital on deforestation, we adjoin a hypothetical variable $\tilde{S}_i$ that replaces the S_i-input of the deforestation equation while leaving every other equation in the system unchanged. The hypothetical model is

$$
\begin{aligned}
\Delta B_j &= g^{\Delta B_j}(\epsilon_{B_j}) \\
\Delta B_i &= g^{\Delta B_i}(\Delta B_j,\, V_B,\, \epsilon_{B_i}) \\
S_i &= g^{S}(\Delta B_i,\, V_B,\, V_D,\, \epsilon_S) \\
\tilde{S}_i &\sim \text{external, independent of all } \epsilon\text{'s and confounders} \\
D_i &= g^{D}(\tilde{S}_i,\, \Delta B_i,\, V_D,\, \epsilon_D).
\end{aligned}
\tag{A2.5}
$$

Three causal effects are defined in this model. The effect of social capital, β_D^S, is the change in $E[D \mid \tilde{S}, \Delta B_i]$ when $\tilde{S}$ moves with branch presence unchanged. The mediation outside social capital, τ, is the change in D when ΔB_i moves with $\tilde{S}$ held fixed; since branches cannot clear forest themselves, it is the mediation that does not run through social capital. The total effect, β_D^B, is defined by adjoining $\tilde{B}_i$ that replaces ΔB_i in both the social capital and deforestation equations, and in the linear model it equals $\tau + \beta_S^B \beta_D^S$.

Identifying assumptions

We require three assumptions. The first is the exclusion restriction of Section 5; the second is the substantive new restriction that the decomposition depends on; the third is the functional form approximation we use in estimation.

(A1) Outer-ring instrument exclusion. The outer-ring branch change ΔB_j affects D_i and S_i only through its effect on ΔB_i. This is the standard IV exclusion restriction defended in Section 5 and probed in the falsification exercises in Section 6. Under (A1), the total effect of ΔB_i on D_i and the effect of ΔB_i on S_i are identified by IV.

(A2) Mediated confounding. $V_B \perp\!\!\!\perp V_D$, where V_B is the part of the unobserved confounders that drives ΔB_i (and may also move S_i) and V_D is the part that drives D_i (and may also move S_i). This is the mediated confounding assumption which Dippel et al. (2021) introduce for the linear model. Its implication is the conditional exogeneity restriction

$$
D_i(\tilde{s}) \perp\!\!\!\perp \Delta B_j \mid \Delta B_i,
$$

together with the exogeneity of ΔB_i in the deforestation equation. Together they allow ΔB_j to instrument social capital in the deforestation equation while ΔB_i enters as a control. Relevance requires that V_B moves social capital: in the linear model, the partial effect of ΔB_j on S_i given ΔB_i is proportional to $-\lambda_B \pi$, where λ_B is the loading of S_i on V_B and π the first-stage coefficient.

The economic content of (A2) is that the unobserved drivers of bank-network expansion (federal branching strategies of state-owned banks, regional commodity cycles, banks' internal portfolio decisions, the digitization wave) do not act on deforestation inside the Indigenous land except through mediation, while the unobserved drivers of deforestation (shifts in local norms, weather shocks affecting agricultural opportunity costs, changes in criminal-organization activity) do not drive branch placement. Two features of our setting make (A2) credible:

1. The variation we exploit reflects bank decisions made for reasons external to the protected territory: federal mandates on government-owned banks throughout the 2000s and 2010s and the subsequent contraction as digital banking rose. These drivers are network-wide and operate on horizons (a national branching plan; a sectoral digitization wave) much longer than the ILTF-specific shocks that move deforestation inside an Indigenous land.
2. ILTF and year fixed effects absorb persistent ILTF-specific characteristics and common annual shocks. Conditional on these, the residual variation in ΔB_i is more plausibly independent of V_D than the unconditional level.

Two violations deserve mention. A local shock that both attracts branches and changes clearing directly, such as an illegal gold boom that raises demand for cash, would place V_B in the deforestation equation; regional fixed effects in Section 6.2 absorb booms at the state and mesoregion level. Mediation outside social capital that is itself driven by V_B would do the same. (A2) is not directly testable. The placebo regressions on past deforestation in Table 5 support (A1) but do not test (A2).

(A3) Linear approximation. We approximate g^D and g^S by linear functions in their arguments. The model above involves indicator functions and Cobb-Douglas production, so (A3) is an approximation of the local response of aggregate deforestation to marginal shifts in social capital and branch presence around the equilibrium of the structural model. We aggregate the individual-level decision in $\mathbf{1}[U_i > \underline{U}_i]$ to the ILTF level by interpreting D_i as the share of marginal perpetrators above the reservation utility, scaled by the ILTF-specific production possibility frontier; this aggregate is locally linear in its arguments around any interior point.

Proof of Proposition 5.1

Substitute both groups of mediators into the outcome equation. What is left is an equation in S and ΔB only: the coefficient on S picks up every path that starts at S (directly, and through mediators that respond to S), and the coefficient on ΔB picks up every path from branches that skips S (through mediators that respond to branches, including those that also feed into S) plus any direct effect. The IV regression of Equation 6 recovers exactly these two composite coefficients, because Z and ΔB are both uncorrelated

with the composite error (exclusion and mediated confounding) and Z still moves S once ΔB is held fixed (relevance). The first leg picks up every path from branches into S, including through mediators that cause S. Paths into and out of S multiply, so the indirect effect is the sum over all paths through S. Two-way loops between S and the same mediator are excluded because they would make the system simultaneous.

All moments below are population covariances of variables net of the fixed effects and controls; by the Frisch-Waugh-Lovell theorem, the IV estimands of Equation 6 with those controls equal the estimands of the same IV regression on the partialled-out variables.

Proof. The argument extends the identification result of Dippel et al. (2021) to a model with further mediators. Substituting the equations for N_j and then M_k into the equation for D gives

$$D \;=\; b^{*} S \;+\; t^{*}\,\Delta B \;+\; V,$$

$$b^{*} \equiv \delta + \sum_{j} \phi_j \rho_j, \qquad t^{*} \equiv \gamma + \sum_{k} \theta_k \alpha_k + \sum_{j} \phi_j a_j, \qquad V \equiv U_D + \sum_{k} \theta_k U_k + \sum_{j} \phi_j V_j.$$

The IV estimands (b, t) of Equation 6 solve the moment conditions $\mathrm{Cov}(W, D - bS - t\,\Delta B) = 0$ for the two instruments $W \in \{Z, \Delta B\}$:

$$\mathrm{Cov}(Z, D) = b\,\mathrm{Cov}(Z, S) + t\,\mathrm{Cov}(Z, \Delta B), \qquad \mathrm{Cov}(\Delta B, D) = b\,\mathrm{Cov}(\Delta B, S) + t\,\mathrm{Var}(\Delta B).$$

From the substituted equation, $\mathrm{Cov}(Z, D) = b^{*}\mathrm{Cov}(Z, S) + t^{*}\mathrm{Cov}(Z, \Delta B) + \mathrm{Cov}(Z, V)$ and $\mathrm{Cov}(\Delta B, D) = b^{*}\mathrm{Cov}(\Delta B, S) + t^{*}\mathrm{Var}(\Delta B) + \mathrm{Cov}(\Delta B, V)$. Condition (i) gives $\mathrm{Cov}(Z, V) = 0$ and condition (ii) gives $\mathrm{Cov}(\Delta B, V) = 0$. Subtracting the moment conditions,

$$\begin{pmatrix} \mathrm{Cov}(Z, S) & \mathrm{Cov}(Z, \Delta B) \\ \mathrm{Cov}(\Delta B, S) & \mathrm{Var}(\Delta B) \end{pmatrix} \begin{pmatrix} b - b^{*} \\ t - t^{*} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The determinant of the matrix is non-zero by condition (iii), so $b = b^{*}$ and $t = t^{*}$. For the first leg, condition (i) gives $\mathrm{Cov}(Z, M_k) = \alpha_k\,\mathrm{Cov}(Z, \Delta B)$ and

$$\mathrm{Cov}(Z, S) \;=\; \psi\,\mathrm{Cov}(Z, \Delta B) + \sum_{k} \pi_k\,\mathrm{Cov}(Z, M_k) + \mathrm{Cov}(Z, U_S) \;=\; \Big(\psi + \sum_{k} \pi_k \alpha_k\Big)\,\mathrm{Cov}(Z, \Delta B),$$

so the IV estimand of Equation 5 is $\beta^{B}_{S} = \psi + \sum_k \pi_k \alpha_k$. For the total effect, $\beta^{B}_{D} = \mathrm{Cov}(Z, D)/\mathrm{Cov}(Z, \Delta B)$, with $\mathrm{Cov}(Z, \Delta B) \neq 0$ by (iii). Using $\mathrm{Cov}(Z, V) = 0$ and the expression for $\mathrm{Cov}(Z, S)$,

$$\beta^{B}_{D} \;=\; \frac{b^{*}\beta^{B}_{S}\,\mathrm{Cov}(Z, \Delta B) + t^{*}\,\mathrm{Cov}(Z, \Delta B)}{\mathrm{Cov}(Z, \Delta B)} \;=\; \beta^{B}_{S}\,\beta^{S}_{D} + \tau.$$

Setting $\gamma = 0$ gives $\tau = \sum_k \theta_k \alpha_k + \sum_j \phi_j a_j$. □

Appendix C: The social capital index and the channel estimates

Joint-GMM estimates, sensitivity and weak-instrument sets

The decomposition in Figure 7 is estimated jointly by GMM. We stack the moment conditions of Equation 5 and Equation 6, so that the effect of an additional branch on the social capital index (b_1), the effect of the instrumented index on cumulative deforestation (b_2) and the mediation outside social capital (τ) are estimated together with a cluster-robust covariance. In Equation 6, $\Delta B_{i,t}$ is treated as exogenous, as mediated confounding implies. Table A3.2 reports these estimates by horizon. The indirect effect is $b_1 b_2$, and its standard error follows from the delta method using the full covariance of (b_1, b_2) rather than treating the two legs as independent. The total effect is the baseline IV estimate on the same sample, and it equals the indirect effect plus τ. The estimation is exactly identified, so no overidentification test applies.

Table A3.1 reports the two legs estimated equation by equation, each on its own available sample, including the first-stage statistics and the coefficient on $\Delta B_{i,t}$ in the second leg. The joint GMM of Table A3.2 uses the sample common to both legs, which is why the first leg differs between the two tables.

Dependent Variables:	Social Capital Index	1 yr	2 yr	3 yr	4 yr	5 yr
Variables						
Δ Branches	0.298***	0.289	0.330	0.306	0.245	-0.043
	(0.103)	(0.198)	(0.356)	(0.466)	(0.563)	(0.470)
Social Capital Index		-2.742*	-5.611*	-7.283*	-6.963	-4.894*
		(1.645)	(3.120)	(4.161)	(4.384)	(2.952)
Deforest. Indigenous land (t-1)	0.005	0.195***	0.348*	0.244	0.063	-0.276
	(0.004)	(0.057)	(0.178)	(0.221)	(0.178)	(0.339)
Deforest. non-protected areas (t-1)	-0.001**	-0.003*	-0.015*	-0.024*	-0.029	-0.030
	(0.000)	(0.002)	(0.008)	(0.014)	(0.019)	(0.019)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	5,952	5,580	5,208	4,836	4,464	4,092
R^2	0.388	0.407	0.493	0.547	0.591	0.647
Wald (1st stage), Δ Branches	34.853					
Wald (1st stage), Social Capital Index		13.210	15.822	17.473	21.055	41.883

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A3.1: Mediation of the effect of bank presence on Indigenous land deforestation through social capital.

Table A3.3 repeats the decomposition under the mesoregion-by-year fixed effects of Section 6.2. The mediation through social capital survives the saturation: it is larger than in the headline specification and significant at the 5 percent level at all five horizons, against three in Table A3.2. The mediation outside

			Through social capital		Outside social capital			First-stage Wald	
Horizon	β_S^B	β_D^S	Effect	SE	τ	SE	Total	ΔB	S
T+1	0.365	-2.742	-1.000	0.491	0.289	0.198	-0.711	39.2	13.2
T+2	0.395	-5.611	-2.214	1.020	0.330	0.356	-1.884	38.5	15.8
T+3	0.418	-7.283	-3.047	1.521	0.306	0.466	-2.741	37.2	17.5
T+4	0.429	-6.963	-2.989	1.726	0.245	0.562	-2.744	40.8	21.1
T+5	0.524	-4.894	-2.565	1.442	-0.043	0.469	-2.608	44.3	41.9

Table A3.2: Joint-GMM decomposition of the effect of one additional branch on cumulative deforestation, by horizon. Mediator: social capital index. The indirect effect is the product of the two legs, with a delta-method standard error from the full cluster-robust covariance. τ is the mediation outside social capital. Year and ILTF fixed effects.

			Through social capital		Outside social capital			First-stage Wald	
Horizon	β_S^B	β_D^S	Effect	SE	τ	SE	Total	ΔB	S
T+1	0.371	-4.242	-1.572	0.670	0.445	0.290	-1.127	30.7	16.0
T+2	0.369	-8.279	-3.054	1.231	0.517	0.471	-2.537	30.7	16.0
T+3	0.360	-11.078	-3.983	1.722	0.461	0.573	-3.522	30.1	18.6
T+4	0.391	-10.836	-4.237	1.990	0.462	0.646	-3.775	30.1	18.6
T+5	0.538	-8.037	-4.325	1.880	0.078	0.553	-4.247	28.3	18.2

Table A3.3: Joint-GMM decomposition under mesoregion-by-year fixed effects. Columns as in Table A3.2.

social capital remains indistinguishable from zero at every horizon, with a largest absolute t-statistic of 1.54. As for the baseline, the first-stage statistics fall when regional variation is absorbed, and remain above conventional thresholds.

Table A3.4 reports, for each horizon, the violation of mediated confounding at which the indirect effect through social capital falls to zero. Let V be the composite error of the deforestation equation in Proposition 5.1. If $\mathrm{Cov}(\Delta B, V) = \kappa \,\mathrm{Var}(\Delta B)$ instead of zero, the second moment condition of Equation 6 shifts by $\kappa \,\mathrm{Var}(\Delta B)$. Solving the system in the proof of Proposition 5.1 with this shift gives $b(\kappa) = b^* + \kappa \,\mathrm{Cov}(Z, \Delta B)\,\mathrm{Var}(\Delta B)/\det$, where det is the determinant in condition (iii), while the total effect is unaffected. The table reports the value κ^0 at which $b(\kappa) = 0$, in km^2 per branch and relative to the total effect.

Horizon	Indirect	Total	κ^0	$\lvert\kappa^0\rvert$ / |total|
T+1	-1.00	-0.71	0.75	1.06
T+2	-2.21	-1.88	1.73	0.92
T+3	-3.05	-2.74	2.38	0.87
T+4	-2.99	-2.74	2.29	0.84
T+5	-2.57	-2.61	2.11	0.81

Table A3.4: Sensitivity of the mediation through social capital to violations of mediated confounding. κ^0 is the correlation between the branch change and the unobserved determinants of deforestation, in km^2 per branch, at which the indirect effect through social capital falls to zero.

Weak instruments in either leg would make the delta-method intervals unreliable. Anderson-Rubin confidence sets (Anderson and Rubin, 1949) remain valid in that case. For the effect of a branch on the social capital index, Equation 5, the 95 percent set is [0.12, 0.53]. For the effect of social capital on cumulative deforestation, Equation 6, the sets at one to five years are [-7.61, -0.13], [-15.57, -0.50], [-21.29, -0.43], [-21.92, 0.73], [-14.77, 0.39]. They exclude zero at 3 of the five horizons, so the effect of social capital on deforestation is robust to weak instruments at short and medium horizons.

Individual social capital metrics

The social capital index combines two inputs. Table A3.5 and Table A3.6 repeat the two legs of Equation 5 and Equation 6 with each input as the mediator: non-profit association employment per 100,000 inhabitants, and the share of valid votes in election years.

Dependent Variables:	Empl. Assoc.	1 yr	2 yr	3 yr	4 yr	5 yr
Variables						
Δ Branches	29.449***	1.742	4.551	4.857	15.535	2.507
	(10.360)	(2.177)	(6.861)	(6.123)	(42.724)	(3.387)
Empl. Assoc.		-0.097	-0.264	-0.271	-0.870	-0.257
		(0.114)	(0.355)	(0.287)	(2.221)	(0.239)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes	Yes
Lagged deforestation	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	6,096	5,715	5,334	4,953	4,572	4,191
R^2	0.848	0.407	0.493	0.548	0.591	0.648
Wald (1st stage), Δ Branches	35.822					
Wald (1st stage), Empl. Assoc.		0.957	0.679	1.246	0.167	2.038

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A3.5: Mediation effect on Indigenous land deforestation through social capital (association employment)

Dependent Variables:	Share of valid votes	1 yr	2 yr	3 yr	4 yr	5 yr
Variables						
Δ Branches	2.028***	0.260	0.256	-0.006	-0.039	-0.250
	(0.442)	(0.209)	(0.398)	(0.489)	(0.535)	(0.585)
Share of valid votes		-0.462	-0.928*	-0.921*	-0.954*	-0.820*
		(0.285)	(0.522)	(0.509)	(0.572)	(0.466)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes	Yes
Lagged deforestation	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	3,048	2,667	2,667	2,286	2,286	1,905
R^2	0.506	0.365	0.453	0.559	0.602	0.684
Wald (1st stage), Δ Branches	43.422					
Wald (1st stage), Share of valid votes		40.189	40.189	44.941	44.941	40.748

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A3.6: Mediation effect on Indigenous land deforestation through social capital (votes)

Social capital index with a common scale

The index in the main text is the first principal component of its two inputs computed within each ILTF, so its units differ across lands. We recompute it as the first principal component of the standardized inputs

over the whole panel, which puts all lands on a common scale, and repeat the decomposition. Table A3.7 reports the results.

Horizon	b_1	b_2	Indirect	τ	Total	Wald, leg 2
T+1	0.196	-3.630	-0.71 (0.36)	-0.00 (0.12)	-0.71 (0.34)	13.6
T+2	0.241	-7.084	-1.70 (0.81)	-0.18 (0.25)	-1.88 (0.77)	16.3
T+3	0.265	-8.996	-2.38 (1.20)	-0.36 (0.35)	-2.74 (1.19)	16.9
T+4	0.350	-6.827	-2.39 (1.39)	0.00 (0.44)	-2.77 (1.33)	20.9
T+5	0.401	-5.838	-2.34 (1.31)	-0.30 (0.46)	-2.64 (1.23)	18.0

Table A3.7: Decomposition with the social capital index computed on a common scale across Indigenous lands. Joint GMM as in Table A3.2; standard errors in parentheses.

With the pooled index, the indirect effect through social capital at three years is -2.38 km^2 (standard error 1.20) and the mediation outside social capital is -0.36 km^2, against -3.05 and 0.31 km^2 with the index of the main text.

The index and other local conditions

A natural concern is that the index tracks local prosperity rather than civic life. Table A3.8 reports its correlation with the main alternatives, raw and after removing Indigenous-land and year fixed effects. The second column is the relevant one: it is the variation the estimates use.

Variable	Raw	Within land	N
log GDP	0.073	-0.019	4,836
GDP growth	-0.011	-0.071	4,836
log exports	-0.073	-0.006	3,026
Rural credit / GDP	0.060	-0.042	4,836
Tax revenue / GDP	0.095	-0.068	4,836
Vehicles per km^2	0.026	0.039	5,952
Population	0.019	-0.044	5,952
Urban expansion	-0.074	-0.043	5,952

Table A3.8: Correlation between the social capital index and alternative local conditions. The within-land column residualises both series on Indigenous-land and year fixed effects, which is the variation the estimates use. Sources: IBGE (gross domestic product, population), Ministry of Development, Industry, Trade and Services (exports), Banco Central do Brasil (rural credit), National Treasury SICONFI (tax revenue), National Traffic Department DENATRAN (vehicle fleet), and the MapBiomas urban module (urban expansion).

No within-land correlation exceeds 0.07 in absolute value.

Appendix D: Which banks drive the effect?

The baseline results reported in Table 3 do not differentiate between the ownership type of banks. Since government-owned banks have responded in the sample period to economic policy rather than deciding their strategy purely for business-oriented decisions, it is plausible that the banking presence dynamics between these two groups are different. Figure A4.1 re-estimates the local-projection IV of Equation 2 separately for federal retail branches (Banco do Brasil and Caixa Econômica) and for private retail branches, each instrumented by the same total outer-ring change, alongside the pooled all-banks baseline.

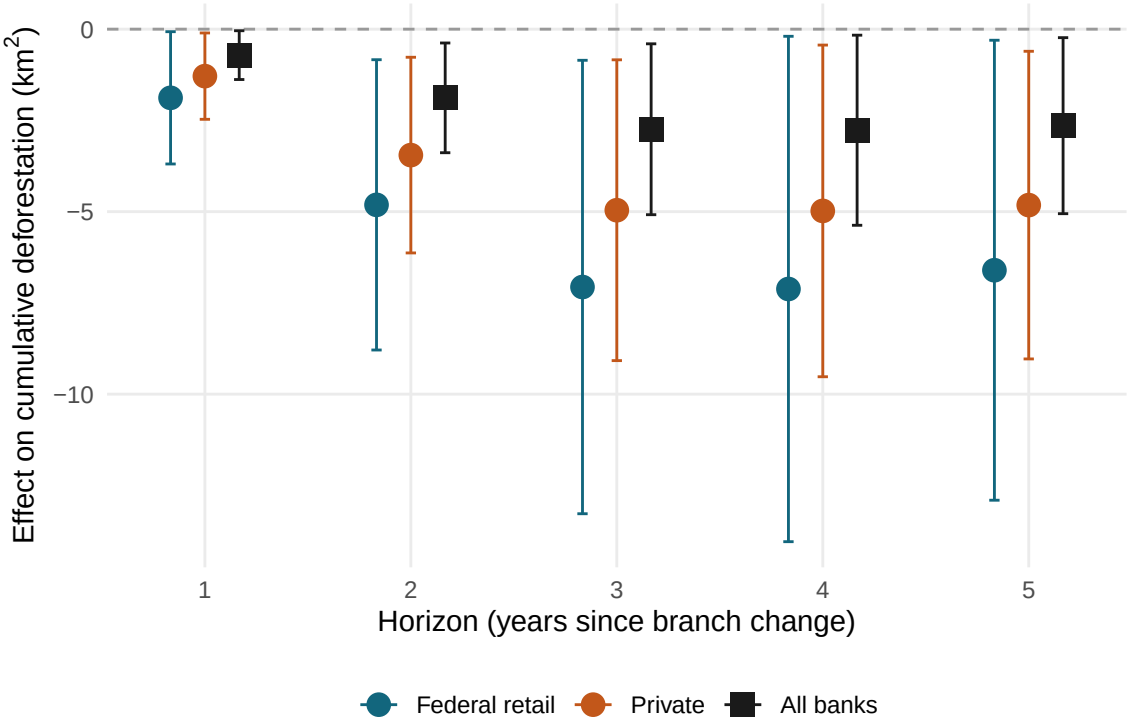


Figure A4.1: Local-projection IV estimates by bank ownership. Each point is the effect of one additional branch on cumulative Indigenous-land deforestation (square kilometres) at the given horizon, estimated separately for federal retail branches (Banco do Brasil, Caixa Econômica) and private retail branches, each instrumented by the total outer-ring change, alongside the pooled all-banks baseline. Bars are 95 percent confidence intervals, clustered by Indigenous land.

Both federal and private retail branches contribute to the protective effect. The estimates are negative and significant at the 5 percent level at all five horizons for federal retail branches and at all five horizons for private retail branches, so the pooled result is not driven by a single ownership type. The remaining institutions, the development banks, the Pará state bank, and non-bank lenders, are too thin at the frontier for the outer-ring instrument to move, so their effect is not separately identified and is omitted here.

Appendix E: Meeting places, from branches to venues to the forest

This subsection completes the chain from bank branches to social capital and ultimately to reduced deforestation inside Indigenous lands. The main text shows in Table 4 that bank branch entry raises the per-capita stock of meeting-place activities. We now show that the same activities mediate the bank-presence effect on deforestation when used directly as the mediator in place of the composite index. Table A5.1 summarizes that exercise across all five categories, reporting alongside each estimate the first-stage Wald statistic for branches on the business and, as a range across horizons, for the business on deforestation. Only restaurants and bars and supermarkets carry a usable first stage into the second step; for hair salons, entertainment and artistic activities the instrument shifts the mediator too little for the horizon estimates to be informative, so their insignificance there is not evidence of no effect. Table A5.2 and Table A5.3 then report the full first and second stages for the two categories that do. In both cases the pattern reproduces Table A3.1: branch entry raises the per-capita stock of these venues, and exogenous variation in the venues lowers cumulative deforestation inside Indigenous lands. The model fits use the same instrument as the headline IV, run on the full panel, and cluster standard errors at the ILTF level.

	Cumulative deforestation after					Wald (1st stage)	
	1 yr	2 yr	3 yr	4 yr	5 yr	Branches	Business
Rest. & bars	-9.50	-18.53*	-19.60*	-23.34	-21.34	38.1	6.5–16.1
	(6.05)	(10.60)	(10.95)	(15.09)	(13.24)		
Supermarkets	-16.46	-33.67	-31.90*	-54.96	-88.53	35.8	1.9–10.9
	(11.55)	(21.10)	(17.90)	(39.63)	(81.56)		
Hair salons	293.35	-171.22	-205.20	-89.16	-38.62	31.0	0.0–1.3
	(6123.91)	(938.19)	(962.52)	(180.71)	(43.01)		
Entertainment	-42.24	-57.35	-77.41	-50.26	-32.61	32.7	0.2–3.0
	(101.74)	(86.62)	(117.25)	(59.99)	(29.92)		
Artistic	-90.53	-134.35	-171.26	-225.28	-1277.30	26.8	0.0–0.1
	(434.18)	(499.43)	(573.94)	(928.40)	(30680.18)		

Notes: Each business is instrumented by the outer-ring branch change. The final columns report the first-stage Wald statistic for branches on the business, both estimated on the sample displayed here, and the range across horizons for the business on deforestation. Year and ILTF fixed effects and lagged deforestation controls throughout. Clustered (ILTF) standard errors in parentheses. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.

Table A5.1: Local businesses and cumulative deforestation.

Dependent Variables:	Rest. & bars	1 yr	2 yr	3 yr	4 yr	5 yr
Variables						
Δ Branches	0.104***	0.172	0.088	0.022	0.354	0.454
	(0.040)	(0.171)	(0.322)	(0.388)	(0.656)	(0.731)
Rest. & bars		-9.499	-18.528*	-19.597*	-23.341	-21.341
		(6.053)	(10.603)	(10.946)	(15.090)	(13.242)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes	Yes
Lagged deforestation	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	5,543	5,163	4,789	4,433	4,116	3,802
R^2	0.904	0.406	0.493	0.547	0.590	0.647
Wald (1st stage), Δ Branches	38.067					
Wald (1st stage), Rest. & bars		6.509	9.636	16.102	12.984	16.078

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A5.2: Mediation of the bank-presence effect through restaurants and bars, measured per 100,000 inhabitants.

Dependent Variables:	Supermarkets	1 yr	2 yr	3 yr	4 yr	5 yr
Variables						
Δ Branches	0.073**	0.329	0.337	0.420	1.182	2.133
	(0.029)	(0.285)	(0.495)	(0.585)	(1.303)	(2.530)
Supermarkets		-16.460	-33.674	-31.904*	-54.959	-88.530
		(11.552)	(21.098)	(17.899)	(39.633)	(81.556)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
ILTF FE	Yes	Yes	Yes	Yes	Yes	Yes
Lagged deforestation	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	6,096	5,715	5,334	4,953	4,572	4,191
R^2	0.801	0.407	0.493	0.548	0.591	0.648
Wald (1st stage), Δ Branches	35.822					
Wald (1st stage), Supermarkets		3.654	5.038	10.863	4.659	1.866

Clustered (ILTF) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A5.3: Mediation of the bank-presence effect through supermarkets, measured per 100,000 inhabitants.

Appendix F: An instrument from banks' network expansion

The outer-ring instrument uses the geography around each land. This appendix builds an instrument that does not. Branch changes in the Amazon are driven to a large extent by the network decisions of a few banks, including federal programs that directed government-owned banks to underbanked towns. Let $B_{i,b,2007}$ be the number of branches of bank b in the ILTF of land i in 2007, the year before the sample starts, and let $g_{b,t}^{-i}$ be the growth rate of bank b's branch network in the BLA in year t, excluding the municipalities in the ILTF of land i and in its outer ring. The instrument is

$$Z_{i,t}^{SS} = \sum_{b} B_{i,b,2007}\, g_{b,t}^{-i}, \tag{A6.1}$$

the change in branches the ILTF would have experienced had each incumbent bank grown at its network-wide rate elsewhere. Following Borusyak et al. (2022), identification comes from the bank-level growth rates, which must be unrelated to the unobserved determinants of deforestation in land i, while the 2007 exposures may be endogenous. Because a few banks account for most branches, the effective number of shocks is small, and inference that treats them as many independent draws would be too optimistic; we therefore report standard errors clustered by Indigenous land and use the instrument as a check on the baseline rather than as the main specification.

Table A6.1 reports the estimates. The bank leave-one-out instrument has a first-stage Wald statistic between 7.7 and 22.7 across horizons, and gives -3.60 km^2 (standard error 2.02) at three years, against -2.74 km^2 with the outer ring. Up to three years, the over-identification test does not reject that the two instruments identify the same effect (the smallest p-value is 0.45). At longer horizons the leave-one-out instrument weakens and the test rejects, so we do not use it to measure the long-run effect.

	Cumulative deforestation inside Indigenous lands				
	1 yr	2 yr	3 yr	4 yr	5 yr
Instrument					
Outer ring (baseline)	-0.711^{**}	-1.882^{**}	-2.742^{**}	-2.769^{**}	-2.644^{**}
	(0.341)	(0.767)	(1.194)	(1.329)	(1.230)
Wald (1st stage)	40.2	39.6	38.3	41.9	45.6
Bank leave-one-out	-0.776	-1.974	-3.599^{*}	-6.327^{*}	-9.663^{*}
	(0.561)	(1.294)	(2.018)	(3.515)	(5.444)
Wald (1st stage)	22.7	20.4	17.9	10.9	7.7
Outer ring and bank leave-one-out	-0.726^{**}	-1.903^{**}	-2.935^{**}	-3.376^{**}	-3.526^{**}
	(0.364)	(0.837)	(1.304)	(1.547)	(1.517)
Wald (1st stage)	26.2	24.9	23.5	22.9	24.5
Sargan test, p-value	0.88	0.91	0.45	0.03	0.00
Observations	5,715	5,334	4,953	4,572	4,191

ILTF and year fixed effects and lagged deforestation controls throughout
Standard errors clustered by Indigenous land
*Signif. codes: ***: 0.01, **: 0.05, *: 0.1*

Table A6.1: Local-projection IV estimates with the bank leave-one-out instrument.